

Non-Abelian gauge theories

1. Recap of gauge theory and brief look at Lie group theory

1.1 Recap: Abelian gauge theory

We firstly start with QED, which as we know is an Abelian $U(1)$ gauge theory. It start from the free fermion Lagrangian:

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi. \quad (3.1)$$

The Noether current is: $j^\mu = \bar{\psi}\gamma^\mu\psi$. To approach the QED Lagrangian, we introduce the (global) gauge transformation as:

$$\psi \rightarrow \psi' = e^{i\alpha}\psi \quad (3.2)$$

$$\bar{\psi} \rightarrow \bar{\psi}' = e^{-i\alpha}\bar{\psi} \quad , \quad \alpha \in \mathbb{R}. \quad (3.3)$$

The local gauge transformation is just changing the parameter α rely on the coordinate:

$$\alpha \rightarrow \alpha(x). \quad (3.4)$$

The gauge symmetry is saying that the Lagrangian is invariant under the gauge transformation. This requires us to introduce the covariant derivative D_μ :

$$D_\mu\psi(x) = (\partial_\mu - igA_\mu)\psi(x). \quad (3.5)$$

where in QED case, the charge $g = e$, and the gauge field A_μ is just the photon field, which satisfies the transformation rule:

$$A_\mu \rightarrow A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha(x). \quad (3.6)$$

Then we immediately have the covariant derivative transfers as

$$D_\mu\psi \rightarrow (D_\mu\psi)' = \partial_\mu\psi' - igA'_\mu\psi' = e^{i\alpha}D_\mu\psi. \quad (3.7)$$

Thus the invariant of $\mathcal{L}$ requires that

$$\mathcal{L}(\psi, A_\mu) = \mathcal{L}(\psi', A'_\mu) = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi \quad (3.8)$$

$$= i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi + gA_\mu j^\mu. \quad (3.9)$$

For photon dynamical, we need to add kinetic term for A_μ :

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (3.10)$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1.2 $SU(N)$ gauge theory

Now we consider the non-Abelian $SU(N)$ gauge theory with N Dirac spinors ψ_j , where $j = 1, \dots, N$. The original Lagrangian is

$$\mathcal{L} = \sum_{j=1}^N i\bar{\psi}_j \gamma^\mu \partial_\mu \psi_j - m\bar{\psi}_j \psi_j. \quad (3.11)$$

Under the global unitary transformation $U_{ij} \in \mathbb{C}$, $UU^\dagger = \mathbb{1}$, the spinors obey

$$\psi_j \rightarrow \psi'_j = \sum_k U_{jk} \psi_k, \quad (3.12)$$

$$\bar{\psi}_j \rightarrow \bar{\psi}'_j = \sum_k \bar{\psi}_k U_{kj}^\dagger, \quad (3.13)$$

thus the mass term transformation under $SU(N)$

$$\sum_j \bar{\psi}_j \psi_j \rightarrow \sum_j \bar{\psi}'_j \psi'_j = \sum_{k,l,j} \bar{\psi}_k \underbrace{U_{kj}^\dagger U_{jl}}_{\delta_{kl}} \psi_l = \sum_k \bar{\psi}_k \psi_k \quad (3.14)$$

is invariant.

Similar with the Abelian case, the local gauge transformation is $U \rightarrow U(x)$. We also require the Lagrangian invariant under the transformation. This gives coupling of fermions to gauge field in covariant derivative analogous to QED:

$$D_\mu \psi_j = \partial_\mu \psi_j - ig \sum_k (A_\mu)_{jk} \psi_k. \quad (3.15)$$

This form of covariant derivative also requires the transformation rule as

$$D_\mu \psi_j \rightarrow (D_\mu \psi_j)' = \partial_\mu \psi'_j - ig \sum_k (A'_\mu)_{jk} \psi'_k \quad (3.16)$$

$$= \sum_k U_{jk} (D_\mu \psi_k) \quad (3.17)$$

For convenient, we will use the short-hand notation (Matrix notation), which basically just ignore the index of the matrix element:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = U D_\mu \psi = e^{i\alpha(x)} D_\mu \psi. \quad (3.18)$$

We claim that the gauge field transforms following

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.19)$$

Although we can directly use the definition to compute this expression, I just don't want to write so much. Instead, we can have a quick check that this claim indeed satisfies our requirement:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = \partial_\mu (U\psi) - ig \left(U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger \right) U\psi \quad (3.20)$$

$$= U \partial_\mu \psi + (\partial_\mu U) \psi - ig U A_\mu \psi - (\partial_\mu U) \psi \quad (3.21)$$

$$= U (\partial_\mu \psi - ig A_\mu \psi) = U D_\mu \psi. \quad (3.22)$$

Thus, the Lagrangian is invariant under $SU(N)$ gauge transformation.

Now we need to approach the dynamical of non-Abelian gauge field, which requires a field strength. Since we keep calling it as “non-Abelian”, but until now it’s not quite obvious to see how it is “non-Abelian”, therefore, the field strength should give you some direct idea when we claim:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \quad (3.23)$$

We can also check it is true that this field strength indeed satisfies our requirement, namely keep the Lagrangian invariant. To show this, we firstly have the transformation rule:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu - ig[A'_\mu, A'_\nu] = U F_{\mu\nu} U^\dagger. \quad (3.24)$$

The Lagrangian is proportion to the trace over $F_{\mu\nu} F^{\mu\nu} = (F_{\mu\nu})_{jk} (F^{\mu\nu})^{kj} = \text{tr}(F_{\mu\nu} F^{\mu\nu})$, due to the Lagrangian is a scalar, thus the only possible combination is the trace over $F_{\mu\nu}$. Thus $\text{tr} F_{\mu\nu}$ transfers as

$$\begin{aligned} \text{tr}(F_{\mu\nu} F^{\mu\nu}) &\rightarrow \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) = \text{tr}(U F_{\mu\nu} U^\dagger U F^{\mu\nu} U^\dagger) \\ &= \text{tr}(U U^\dagger F_{\mu\nu} F^{\mu\nu}) \\ &= \text{tr}(F_{\mu\nu} F^{\mu\nu}). \end{aligned} \quad (3.25)$$

Then we get **the Lagrangian of Yang-Mills theory**

$$\mathcal{L} = -\kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}) + \sum_{j,k=1}^N [\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m \bar{\psi}_j \psi_j], \quad (3.26)$$

where κ is normalized constant.

1.3 Recap for essential Lie group theory

To continue our journey, it’s necessary to master some basic knowledge about Lie group and Lie algebra. So in the next, we will recall some essential knowledge about these mathematical tools and show how are they used in gauge theory.

Firstly, we recall the infinitesimal transformation of the gauge operation. For Abelian theory, e.g. QED:

$$e^{i\alpha} \simeq 1 + i\alpha. \quad (3.27)$$

For non-Abelian gauge theory, e.g. Yang Mills theory:

$$U = \mathbb{1} + \hat{H} + \mathcal{O}(\hat{H}^2), \quad (3.28)$$

$$U^\dagger = \mathbb{1} + \hat{H}^\dagger + \mathcal{O}(\hat{H}^2). \quad (3.29)$$

Thus we have

$$U^{-1} = U^\dagger \implies \hat{H} = -\hat{H}^\dagger \quad (\text{anti-hermitian}) \quad (3.30)$$

where we note $\hat{H} = iH$ with $H = H^\dagger$. Expand $\hat{H}$ in parameters of transformation α^a

$$\hat{H}_{jk} = i \sum_a \alpha^a (t^a)_{jk} \quad (3.31)$$

with $\alpha^a \in \mathbb{R}$, $t^a = (t^a)^\dagger$ is hermitian. Thus t^a span a basis in the space of hermitian $N \times N$ matrices such that we can expand $\hat{H}$ in terms of this basis. t^a are called the generators of gauge transformations.

Generally, the unitary $N \times N$ matrix has N^2 linear independent generators due to t^a is $N \times N$. However, in our case, we require $\det U = 1$, which gives one constraint. So there are $\mathbf{N}^2 - \mathbf{1}$ **generators in $\mathbf{SU}(\mathbf{N})$** . Further more, base on $\det U = 1$ and the Jacobi identity $\det(e^M) = e^{\text{tr}(M)}$, we find

$$\det U = \det(e^{\hat{H}}) = e^{\text{tr} \hat{H}} = 1, \quad (3.32)$$

thus, $\text{tr} \hat{H} = 0$ and implies

$$\text{tr}(t^a) = 0. \quad (3.33)$$

So $\mathbf{t}^a$ are **traceless generators**.

Now we discuss the algebra structure for generators t^a . The Lie algebra, namely the commutator is given by:

$$[t^a, t^b] = i \sum_c f^{abc} t^c. \quad (3.34)$$

where f^{abc} is the structure constant of Lie algebra. The elements of Lie group can be expanded around $\mathbb{1}$.

$$g(\alpha) = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + \frac{1}{2} \sum_{a,b=1}^d \alpha_a \alpha_b T^{ab} + \mathcal{O}(\alpha^3), \quad (3.35)$$

where $T^{ab} = T^{ba}$ is symmetric, which can be expanded in terms of the generator t^a, t^b . Notice $g(0) = \mathbb{1}$ and because of $g(\alpha)\mathbb{1} = \mathbb{1}g(\alpha)$, we then have

$$\gamma_a(\alpha_1, \dots, \alpha_d; 0, \dots, 0) = \gamma_a(0, \dots, 0; \alpha_1, \dots, \alpha_d) = \alpha_a \quad (3.36)$$

where the γ is defined by the group multiplication:

$$g(\alpha_1, \dots, \alpha_d) \cdot g(\beta_1, \dots, \beta_d) = g(\gamma_1, \dots, \gamma_d), \quad (3.37)$$

in which γ is the function of α and β :

$$\gamma_a = \gamma_a(\alpha_1, \dots, \alpha_d; \beta_1, \dots, \beta_d). \quad (3.38)$$

Base on the fact that $\gamma(\alpha, 0) = \alpha$ and $\gamma(0, \beta) = \beta$, then we have only one choice to expand γ_a until the second order to keep these relation, namely:

$$\gamma_a = \alpha_a + \beta_a + \sum_{b,c} C_{bc}^a \alpha_b \beta_c + \dots, \quad (3.39)$$

where C_{bc}^a is a unknown coefficient. Thus, use $g(\alpha)g(\beta) = g(\gamma)$, we have for the left-hand side:

$$g(\alpha)g(\beta) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + \frac{1}{2} \sum_{a,b} [(\alpha_a \alpha_b + \beta_a \beta_b) T^{ab} - 2\alpha^a \beta^b t^a t^b] + \dots, \quad (3.40)$$

And for the right-hand side:

$$g(\gamma) = \mathbb{1} + i \sum_a \gamma_a t^a + \frac{1}{2} \sum_{a,b} \gamma_a \gamma_b T^{ab} + \dots \quad (3.41)$$

Thus use (3.39), then the right-hand side becomes

$$g(\gamma(\alpha, \beta)) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + i \sum_{a,b,c} C_{bc}^a \alpha_b \beta_c t^a + \frac{1}{2} \sum_{a,b} (\alpha_a + \beta_a)(\alpha_b + \beta_b) T^{ab} + \dots \quad (3.42)$$

Compare (3.40) with (3.42), we find

$$-t^a t^b = i \sum_c C_c^{ab} t^c. \quad (3.43)$$

this gives the structure constant:

$$[t^a, t^b] = i \sum_{c=1}^d (C_c^{ba} - C_c^{ab}) t^c \quad (3.44)$$

$$= i \sum_{c=1}^d f^{abc} t^c \quad (3.45)$$

where $f^{abc} = -f^{bac}$, and d is the dimension of Lie algebra.

Examples of Lie groups

- **GL(N)**: Group of $N \times N$ matrices in $\mathbb{R}^N$ with $\det M \neq 0$.
- **U(N)**: Group of unitary $N \times N$ matrices in $\mathbb{C}^N$, satisfies

$$U^\dagger U = U U^\dagger = \mathbb{1} \quad \text{and} \quad \det U \neq 0. \quad (3.46)$$

Infinitesimal version:

$$U = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a. \quad (3.47)$$

with $d = N^2$ is the dimension of $U(N)$ Lie algebra/group.

- **SU(N)** := $\{A \in U(N) | \det A = 1\}$. Dimension of Lie algebra $\mathfrak{su}(N)$ is $d = N^2 - 1$. Generators t^a are traceless

$$\det U = \det \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) = 1 + i \sum_{a=1}^d \alpha_a \text{tr } t^a = 1 \quad (3.48)$$

thus

$$\text{tr } t^a = 0 \quad (3.49)$$

- **O(N)**: group of orthogonal $N \times N$ matrices in $\mathbb{R}^N$.

$$O O^T = O^T O = \mathbb{1} \quad (3.50)$$

Infinitesimal version:

$$O = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \quad (3.51)$$

Dimension of Lie algebra $\mathfrak{o}(N)$ is $\frac{1}{2}N(N-1)$. And notice:

$$O O^T = \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) \left(\mathbb{1} + i \sum_{b=1}^d \alpha_b (t^b)^T \right) = \mathbb{1}, \quad (3.52)$$

thus,

$$\mathbb{1} = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + i \sum_{b=1}^d \alpha_b (t^b)^T + \mathcal{O}(\alpha^2) \quad (3.53)$$

$$= \mathbb{1} + i \sum_{a=1}^d (\alpha_a t^a + \alpha_a (t^a)^T), \quad (3.54)$$

which implies

$$t^a = -(t^a)^T. \quad (3.55)$$

- $\mathbf{SO}(N) := \{A \in O(N) \mid \det A = 1\}$
- $\mathbf{SO}(N, M) = \{\Lambda \in \text{Mat}(N+M, \mathbb{R}) \mid \Lambda^T \eta \Lambda = \eta, \det \Lambda = 1\}$, where

$$\eta = \text{diag}(\underbrace{1, \dots, 1}_N, \underbrace{-1, \dots, -1}_M) \quad (3.56)$$

- **Lorentz group:** $SO(1, 3)$

1.4 Representations of Lie algebra

Def: Representation of group

A **representation of a group G on a d_r -dimensional vector space V** is a group homomorphism

$$\rho : G \rightarrow GL(V), \quad (3.57)$$

which maps each group element $g \in G$ to an invertible linear operator $\rho(g)$ acting on V , such that:

$$\rho(g_i) \cdot \rho(g_j) = \rho(g_i g_j) \quad \forall g_i, g_j \in G. \quad (3.58)$$

The dimension d_r of the vector space V is called **the dimension of the representation**.

- A representation ρ is called **reducible** if there exists a non-trivial proper subspace $U \subset V$ (where $U \neq \{0\}$ and $U \neq V$) that is invariant under the action of all group elements, meaning:

$$\rho(g)u \in U \quad \forall g \in G, \forall u \in U. \quad (3.59)$$

- If no such invariant subspace exists, the representation is called irreducible (irrep).

For a Lie group characterized by continuous parameters α , a representation is a mapping $\rho : g(\alpha) \rightarrow M(\alpha)$, where the matrices (or operators) $M(\alpha)$ preserve the group multiplication:

$$M(\alpha) \cdot M(\beta) = M(\gamma) \quad \text{whenever} \quad g(\alpha) \cdot g(\beta) = g(\gamma). \quad (3.60)$$

In the next, we introduce some useful representations.

- **Fundamental(defining) representation**

For a matrix Lie group, the **fundamental representation** is simply the group of matrices themselves, acting on their natural vector space.

Example: $SU(2)$ fundamental representation

- **Representation Space (V):** A 2-dimensional complex vector space $\mathbb{C}^2$, physically corresponding to a spin-1/2 state

$$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle. \quad (3.61)$$

- **Dimension of the representation:** $d = 2$ (since the matrices are 2×2). Note that the dimension of the $SU(2)$ group/algebra itself is $2^2 - 1 = 3$ (the number of

independent generators). The 3 Hermitian generators for this $d = 2$ fundamental representation are the Pauli matrices:

$$t^a = \frac{1}{2}\sigma^a \quad (a = 1, 2, 3), \quad (3.62)$$

which satisfy the Lie algebra $[t^a, t^b] = i\epsilon^{abc}t^c$, and the Pauli matrix identity:

$$\sigma^k \sigma^l = \delta^{kl} \mathbb{1} + i\epsilon^{klm} \sigma^m \quad (3.63)$$

Physically, we can use these fundamental building blocks to construct higher-dimensional representations via tensor products, such as the 3-dimensional spin-1 triplet state. In fact, $SU(2)$ has spin- j representations with dimension $2j + 1$. For instance, $j = 1/2$ corresponds to the 2-dimensional fundamental representation using 2×2 Pauli matrices.

- **Adjoint representation**

Instead of acting on physical states, the **adjoint representation** is a d -dimensional representation where the group/algebra acts on the Lie algebra space itself (spanned by the generators $\{t^a\}$ as basis vectors). The definition is as follows

Adjoint representation

We first define a automorphism map of group G :

$$\Psi : G \rightarrow \text{Aut}(G) \quad (3.64)$$

$$g \mapsto hgh^{-1} \equiv \Psi_h(g), \quad \forall h \in G, \quad (3.65)$$

where $\text{Aut}(G)$ is the automorphism group of G . Then we can define the derivative of Φ_g at the origin noted as Ad_g :

$$\text{Ad}_g = (d\Phi_g)_e : T_e G \rightarrow T_e G, \quad (3.66)$$

where $T_e G$ is the tangent space in the natural element of G . Thus the **adjoint representation** is a homomorphism:

$$\text{Ad} : G \rightarrow \text{Aut}(T_e G) \quad (3.67)$$

$$g \mapsto \text{Ad}_g. \quad (3.68)$$

To obtain the explicit matrix form of the adjoint representation, we consider an infinitesimal group element $g = e + i\alpha^a t^a \in G$. Acting on a generator $t^b \in T_e G$ via the adjoint map Ad_g , we expand to the linear order of α :

$$\text{Ad}_{e+i\alpha^a t^a}(t^b) = (e + i\alpha^a t^a)t^b(e - i\alpha^a t^a) \simeq t^b + i\alpha^a [t^a, t^b] + \mathcal{O}(\alpha^2) \quad (3.69)$$

This motivates us to define the infinitesimal adjoint map $\text{ad}_{t^a} \equiv d(\text{Ad})_e(t^a)$ on the Lie algebra, which acts purely as a commutator:

$$\text{ad}_{t^a}(t^b) = \lim_{\alpha^a \rightarrow 0} \frac{(\text{Ad}_{e+i\alpha^a t^a}(t^b))_{kl} - (t^b)_{kl}}{i\alpha^a} = [t^a, t^b]_{kl} = i \sum_{c=1}^d f^{abc}(t^c)_{kl}. \quad (3.70)$$

Viewing ad_{t^a} as a linear transformation matrix acting on the basis vector t^b to produce t^c , the expansion coefficients (structure constants) directly define the matrix elements of the adjoint generators:

$$(t_{\text{adj}}^a)_{bc} = i f^{abc} \quad (3.71)$$

Hence, the structure constants f^{abc} are not just algebraic constants, they are precisely **used to construct the adjoint representation**.

In this representation, the group elements act on the Lie algebra itself. The generators are $d \times d$ matrices (where $d = N^2 - 1$), with matrix elements defined by (3.71). We can verify that these matrices satisfy the Lie algebra by using the Jacobi identity:

$$\sum_{A=1}^{d_r} (f^{aAe} f^{bcA} + f^{bAe} f^{caA} + f^{cAe} f^{abA}) = 0. \quad (3.72)$$

Writing the commutation relation

$$[t_{\text{adj}}^a, t_{\text{adj}}^b]_{de} = i \sum_{c=1}^d f^{abc} (t_{\text{adj}}^c)_{de}, \quad (3.73)$$

in terms of structure constants gives:

$$\sum_{A=1}^d [(i f^{aAd})(i f^{bAe}) - (i f^{bAd})(i f^{aAe})] = i \sum_{c=1}^d f^{abc} (i f^{cde}). \quad (3.74)$$

Thus we have:

$$\sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe}) = - \sum_{A=1}^d f^{abA} f^{Ade}. \quad (3.75)$$

Using the anti-symmetry property of structure constant: $f^{abc} = -f^{bac}$, we can reduce to the standard Jacobi identity form:

$$0 = \sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe} + f^{Ade} f^{abA}) \quad (3.76)$$

$$= \sum_{A=1}^d (f^{aAd} f^{beA} + f^{bAd} f^{eaA} + f^{eAd} f^{abA}), \quad (3.77)$$

which is nothing but the Jacobi identity (3.72).

1.5 Inner product of generators

Now since the Lie algebra generators are all matrices, so it's natural to consider the inner product of two matrices. The most natural way is by the trace of matrix product. So we define a matrix D_r^{ab} for generators t_r^a as

$$D_r^{ab} = \text{tr}(t_r^a t_r^b). \quad (3.78)$$

For $a, b = 1, \dots, d = N^2 - 1$. Cyclicity of trace doesn't change the matrix, so it's symmetry and real-valued:

$$D_r^{ab} = D_r^{ba}, \quad (3.79)$$

and also Hermitian:

$$(D_r^{ab})^\dagger = \text{tr}((t_r^a)^\dagger (t_r^b)^\dagger) = D_r^{ab}. \quad (3.80)$$

As we know from linear algebra, for any real-valued symmetry matrix, one can always diagonalize it by a orthogonal transformation. Thus, diagonalization of D_r^{ab} gives

$$\text{tr}(t_r^a t_r^b) = C(r) \delta^{ab}. \quad (3.81)$$

Since the generators are diagonal, then the position of each components in the same t_r are equal, which imply that $C(r)$ is the same for any component of t_r . Thus $C(r)$ is called the **index of representation** or **Dynkin index**.

Example: $SU(N)$

The generators for $SU(2)$ is $t_{\text{fund}}^a = \frac{\sigma^a}{2}$ ($a = 1, 2, 3$), where σ^a is the Pauli matrix:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.82)$$

For the fundamental representation matrix, we can check that

$$\text{tr}(t_r^a t_r^a) = \frac{1}{2}. \quad (3.83)$$

So

$$C(r) = \frac{1}{2}. \quad (3.84)$$

More over, for general case $SU(N)$, $C(r) = \frac{1}{2}$ due to the normalization of N -dim matrices are all similar with $SU(2)$. Using the index of representation, we can find the structure constants of $SU(N)$ satisfy:

$$\begin{aligned} \text{tr}([t^a, t^b] t_r^c) &= i f^{abc} \text{tr}(t_r^c t_r^c) \\ &= \text{tr}(t_r^a t_r^b t_r^c - t_r^b t_r^a t_r^c) = i f^{abc} C(r) \\ &= \text{tr}(t_r^a [t_r^b, t_r^c]) = i f^{abc} C(r). \end{aligned} \quad (3.85)$$

This shows that f^{abc} are totally anti-symmetric:

$$f^{abc} = -f^{bac} = f^{bca}. \quad (3.86)$$

As we all learned from quantum mechanics: $[J^2, J_i] = 0$. So similarly, one can define an operator T^2 that satisfies the similar commutation relation. This operator is called the **quadratic Casimir operator**, which is defined as:

$$T_{ij}^2 = \sum_{a=1}^{d(G)} \sum_{k=1}^{d(r)} t_{ik}^a t_{kj}^a, \quad (3.87)$$

where t_{ij}^a is fundamental representation. We can check that for any a , we all have: $[t^a, T^2] = 0$, namely T^2 commutes with all generators:

$$[t^a, T^2] = \left[t^a, \sum_b t^b t^b \right] = \sum_b ([t^a, t^b] t^b + t^b [t^a, t^b]) \quad (3.88)$$

$$= \sum_b \left(\left(i \sum_c f^{abc} t^c \right) t^b + t^b \left(i \sum_c f^{abc} t^c \right) \right) \quad (3.89)$$

$$= i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.90)$$

$$= i \sum_{c,b} f^{acb} (t^b t^c + t^c t^b) \quad (3.91)$$

$$= i \sum_{b,c} (-f^{abc}) (t^c t^b + t^b t^c) \quad (3.92)$$

$$= -i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.93)$$

$$= 0 \quad (3.94)$$

This implies that T^2 must be proportional to identity:

$$T_{ij}^2 = C_2(r) \delta_{ij} \quad (3.95)$$

And $C_2(r)$ is called the **quadratic Casimir coefficient**. It can be obtained by taking the trace in both sides:

$$\text{tr}(T^2) = \text{tr} \left(\sum_{a=1}^{d(G)} t^a t^a \right) = \sum_{a=1}^{d(G)} \text{tr}(t^a t^a) = C_2(r) \text{tr} \delta_{ij}, \quad (3.96)$$

thus we have:

$$C(r) d(G) = C_2(r) d(r) \quad (3.97)$$

1.6 Representation of Lorentz group

The Lorentz group, as we mentioned before, is $SO(1, 3)$ with metric $\eta = (1, -1, -1, -1)$. The group element Λ with infinitesimal expansion can be written as

$$\Lambda_\nu^\mu = \delta_\nu^\mu + i \sum_a \omega^a (t^a)_\nu^\mu + \mathcal{O}(\omega^2). \quad (3.98)$$

Defining relation of $SO(N, M)$ groups says

$$\Lambda^T \eta \Lambda = \eta, \quad (3.99)$$

thus the infinitesimal form implies

$$(\mathbb{1} + i\omega^a (t^a)^T) \cdot \eta \cdot (\mathbb{1} + i\omega^a t^a) = \eta, \quad (3.100)$$

after dropping the $\mathcal{O}(\omega^2)$, one obtains the anti-symmetric relation:

$$(t^a)^T \eta = -\eta t^a. \quad (3.101)$$

As we mentioned in the last subsection, $SO(1, 3)$, as a subgroup of $O(1, 3)$, has $\frac{1}{2} \times 4 \times (4-1) = 6$ generators, physically are 3 rotation + 3 boosts, which usually noted as $J^{\mu\nu}$. To limit the number of free parameters in $J^{\mu\nu}$, we require

$$J^{\rho\sigma} = -J^{\sigma\rho}. \quad (3.102)$$

So we change the note as

$$\sum_a \omega^a t^a \rightarrow \sum_{\rho\sigma} \omega_{\rho\sigma} J^{\rho\sigma}, \quad (3.103)$$

where $J^{\rho\sigma}$ fulfill $SO(1, 3)$ algebra. Generally, all $SO(N, M)$ groups have also spinor representations generators

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (3.104)$$

fulfill $SO(1, 3)$ algebra:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}). \quad (3.105)$$

1.7 Integration over gauge group

Since a Lie group is also a manifold, so we can introduce the integration structure based on the group element g and function $f(g)$, the measurement dg is called Haar measure, which include the left and right invariant measure:

$$\text{left invariant measure : } \int (dg)_L f(g) = \int (dg)_L f(g'g); \quad (3.106)$$

$$\text{right invariant measure : } \int (dg)_R f(g) = \int (dg)_R f(gg'), \quad (3.107)$$

where the normalization is $\int dg \, 1 = 1$, and for most of situation the left and right Haar measure are the same $dg = (dg)_L = (dg)_R$. This measurement is just like the normal integral is invariant under the transformation $x \rightarrow x + C$, where C is a constant:

$$\int dx f(x) = \int dx f(x + C). \quad (3.108)$$

To obtain the computable form of the measure, we need explicit realization of dg with metric tensor M_{ij} :

$$M_{ij} \equiv \text{tr}(g^{-1} \partial_i g \cdot g^{-1} \partial_j g), \quad (3.109)$$

where $\partial_i g = \frac{\partial}{\partial \alpha_i} g(\alpha)$. Then we can connect to the normal integral:

$$\int dg f(g) = N \int d\alpha |\det M|^{1/2} f(g(\alpha)), \quad (3.110)$$

in which the normalization is the Jacobi determinant.

Example

- $U(1) = \{e^{i\theta} \mid -\pi < \theta < \pi\}$: For $g(\theta) = e^{i\theta}$ we have

$$M_{\theta\theta} = \text{tr}(g^{-1} \partial_\theta g \cdot g^{-1} \partial_\theta g) = \text{tr}(i \cdot i) = -1. \quad (3.111)$$

And using the fact

$$1 = N \int_{-\pi}^{\pi} d\theta \cdot 1 = N \cdot 2\pi, \quad (3.112)$$

thus $N = \frac{1}{2\pi}$. So the integral is

$$\int dg f(g) = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta f(e^{i\theta}) \quad (3.113)$$

- $SU(2) = \{a_0 + i\vec{\sigma} \cdot \vec{a} \mid a_0^2 + \vec{a}^2 = 1\}$: For the element $g(a)$ we have:

$$\partial_0 g = \frac{\partial g}{\partial a_0} = \mathbb{1}_{2 \times 2} \quad (3.114)$$

$$\partial_k g = \frac{\partial g}{\partial a_k} = i\sigma_k \quad (k = 1, 2, 3) \quad (3.115)$$

Thus, the metric is

$$M = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}, \quad (3.116)$$

therefore, $|\det M|^{1/2} = \sqrt{16} = 4$. The last piece is the normalization factor, for that we need to compute the integral for unity. Since the constrain require a sphere, we then can express the unity by delta function:

$$1 = N \int d^4a \delta(a^2 - 1) = N \int_0^\infty r^3 dr \int d\Omega_3 \delta(r^2 - 1), \quad (3.117)$$

thus $N = \frac{1}{\pi^2}$. And finally:

$$\int dg f(g) = \frac{1}{\pi^2} \int d^4a \delta(a^2 - 1) f(g(a)). \quad (3.118)$$

- More general case, for $SU(N)$, Schur's orthogonality relation requires

$$\int dg g_{ij}(g^{-1})_{kl} = \frac{1}{N} \delta_{il} \delta_{jk}. \quad (3.119)$$

2. Quantization of non-Abelian gauge theory

2.1 Preparation

To quantize the non-Abelian gauge theory, we need to list what do we have now, everything begins with the Lagrangian:

$$\mathcal{L} = \sum_{j=1}^{d(r)} (i\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m\bar{\psi}_j \psi_j) - \kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}), \quad (3.120)$$

where the covariant derivative is defined as:

$$(D_\mu)_{jk} = \partial_\mu \delta_{jk} - ig(A_\mu)_{jk}, \quad (3.121)$$

and the field strength is:

$$(F_{\mu\nu})_{jk} = [D_\mu, D_\nu]_{jk}. \quad (3.122)$$

Under the gauge transformation, the gauge fields obey the rule:

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.123)$$

To work in the perturbation theory, we need to explicit show the colour indices. Thus we expand A_μ in basis of adjoint representation:

$$(A_\mu)_{jk} = \sum_{a=1}^{d(G)} A_\mu^a t_{jk}^a \quad (3.124)$$

The infinitesimal transformation with $U_{jk} = \mathbb{1}_{jk} + i \sum_a \alpha^a (t^a)_{jk}$ then becomes:

$$\begin{aligned} (\delta A_\mu)_{jk} &= (A'_\mu)_{jk} - (A_\mu)_{jk} \\ &= -(A_\mu)_{jk} + \left(\mathbb{1} + i \sum_a \alpha^a (t^a) \right)_{jm} (A_\mu)_{mn} \left(\mathbb{1} - i \sum_b \alpha^b (t^b) \right)_{nk} + \frac{1}{g} \sum_a (\partial_\mu \alpha^a) (t^a)_{jm} \left(\mathbb{1} - i \sum_b \alpha^b (t^b) \right)_{mk} \\ &\simeq i \sum_{a=1}^{d(G)} \alpha^a A_\mu^b [t^a, t^b]_{jk} + \frac{1}{g} \sum_{a=1}^{d(G)} (\partial_\mu \alpha^a) t_{jk}^a + \mathcal{O}(\alpha^2) \\ &= \frac{1}{g} \left[-g \sum_{b,c=1}^{d(G)} f^{abc} A_\mu^c \alpha^b + \partial_\mu \alpha^a \right] t_{jk}^a, \end{aligned} \quad (3.125)$$

thus we have

$$\delta A_\mu^a = \frac{1}{g} (D_\mu \alpha)^a = \frac{1}{g} \sum_{b=1}^{d(G)} (D_\mu)^{ab} \alpha^b. \quad (3.126)$$

where $(D_\mu)^{ab}$ is the covariant derivative in adjoint representation:

$$(D_\mu)^{ab} = \partial_\mu \delta^{ab} - g \sum_{c=1}^{d(G)} f^{abc} A_\mu^c. \quad (3.127)$$

2.2 Quantization by path integral

Now we are ready to start quantization via path integral method. We have already seen that the action $S[A]$ is gauge invariant. But

$$\int \mathcal{D}A e^{-S[A]} \quad (3.128)$$

is not well defined, because $\mathcal{D}A$ counts infinite gauge redundant. And if one integrate two times over the pure Yang-Mills term $F_{\mu\nu}^a F^{\mu\nu a}$, one can see that the momentum $M_{\mu\nu}$ satisfies

$$M_{\mu\nu} = -k^2 \eta_{\mu\nu} + k_\mu k_\nu, \quad (3.129)$$

which is degenerated (or not invertible):

$$M_{\mu\nu} k^\nu = -k^2 k_\mu + k_\mu k^2 = 0, \quad (3.130)$$

which means the determinate is 0. So we cannot define the propagator due to it's nothing but the inverse of the momentum operator, which we proved in chapter 1.

The idea to solve these problem comes from Faddeev and Popov. To divide out redundant gauge degrees of freedom, we require a gauge condition

$$G^a(A^u) = 0, \quad (3.131)$$

for a particular point A^u in an orbit of the gauge group, where u is a group element of the non-Abelian gauge group (e.g. $SU(3)$). Therefore, A^u is a gauge choice for A , in the other words, A is gauge-less. So like the way we having $\int \delta(x-a)dx = 1$, we insert the gauge condition into the delta function and define a similar quantity called **Faddeev-Popov(FP) determinant** $\Delta_G[\mathbf{A}]$:

$$\Delta_G^{-1}[A] \equiv \int \mathcal{D}u \delta(G^a(A^u)). \quad (3.132)$$

So by this definition, one can get a useful identity:

$$1 = \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)). \quad (3.133)$$

One can check that this functional determinant is gauge invariant by continuously doing two times gauge transformation: u and u' . So the fields change to $(A^u)^{u'} = A^{u'u}$. So by definition, the new determinant is given by

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u \delta(G^a((A^{u'})^u)) = \int \mathcal{D}u \delta(G^a(A^{u'u})), \quad (3.134)$$

Now we introduce a new group element $u'' = u'u$, in which transformation the measure keeps invariant: $\mathcal{D}u = \mathcal{D}u''$, thus

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u'' \delta(G^a(A^{u''})). \quad (3.135)$$

Changing the integral variable name from u'' to u , we then verify that the FP determinant is indeed invariant under gauge transformation:

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u \delta(G^a A^u) = \Delta_G^{-1}[A] \quad (3.136)$$

Now we insert (3.133) into path integral

$$\int \mathcal{D}A (1) e^{-S[A]} = \int \mathcal{D}A \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)) e^{-S[A]}. \quad (3.137)$$

Since only the delta function in the middle integral is u dependence, and the measure $\mathcal{D}$, the determinant $\Delta_G[A]$ and the action $S[A]$ are all gauge invariant, so we can change the variable (or choose this specific gauge) $A \rightarrow A' = A^u$ without causing any difference:

$$\int \mathcal{D}A \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)) e^{-S[A]} = \int \mathcal{D}A'^{u^{-1}} \Delta_G[A'^{u^{-1}}] \int \mathcal{D}u \delta(G^a(A')) e^{-S[A'^{u^{-1}}]} \quad (3.138)$$

$$= \int \mathcal{D}A' \Delta_G[A'] \int \mathcal{D}u \delta(G^a(A')) e^{-S[A']}. \quad (3.139)$$

The trick that we are playing now can be understood with a simple comparison in calculus. For an integral in one dimension

$$I = \int_{-\infty}^{+\infty} dx \int_0^{2\pi} d\theta f(x + \theta)g(x + \theta), \quad (3.140)$$

which obviously is θ dependent. But we can change the variable $y \equiv x + \theta$. Since θ in the second integral is a constant, thus $dy = dx$ with the same integration range. So the integral changes to

$$I = \int_{-\infty}^{+\infty} dy \int_0^{2\pi} d\theta f(y)g(y), \quad (3.141)$$

where the integrand is now θ independent. With the same trick, now the functional integral is u independent, so we can change the integration sequence and rename the variable from A' to A as

$$\left(\int \mathcal{D}u \right) \cdot \int \mathcal{D}A \Delta_G[A] \delta(G^a(A)) e^{-S[A]}, \quad (3.142)$$

and since the integral over $\mathcal{D}u$ represents the “volume” of the gauge group, which is infinite. So we can just consider the rest part, which is physical and removed all the gauge redundancy as the correct definition of path integral for non-Abelian gauge theories:

$$\int \mathcal{D}A \Delta_G[A] \delta(G^a(A)) e^{-S[A]}. \quad (3.143)$$

So far so good, it seems that we successfully eliminated the gauge redundancy. However, if we consider a specific example, say Coulomb gauge $G^a(A) = \nabla \cdot A^a = 0$, and consider the derivative of G with respect to α , so we have

$$\frac{\delta G_\mu^a}{\delta \alpha^b} = \nabla \cdot \frac{\delta A_\mu^a}{\delta \alpha^b} = \frac{1}{g} \nabla \cdot D_\mu^{ab}, \quad (3.144)$$

where δA_μ^a is given by (3.126), and (3.127) gives

$$\vec{\nabla} \cdot \vec{D}^{ab} = \nabla^2 \delta^{ab} - g f^{abc} \vec{A}^c \cdot \vec{\nabla}, \quad (3.145)$$

which is the Coulomb gauge $\vec{\nabla} \cdot \vec{D}^{ab}$ operator. The problem is it can have non-trivial solution for A_i , namely,

$$(\vec{\nabla} \cdot \vec{D}^{ab}) \alpha^b = 0, \quad (3.146)$$

where $\alpha^b(x) \neq 0$. It means we find a new field by the gauge transformation $\alpha^b(x)$ as $A' = A + \delta A$, with the new gauge condition:

$$G^a(A') = G^a(A) + \delta G^a = 0 + \frac{1}{g} (\vec{\nabla} \cdot \vec{D}^{ab}) \alpha^b = 0, \quad (3.147)$$

which also satisfies the Coulomb gauge $\vec{\nabla} \cdot \vec{A}'^a = 0$. This problem is called the **Gribov Ambiguity**, which only troubles in non-Abelian gauge theory, e.g. QCD, due to the Abelian gauge theory have the boundary $\alpha \rightarrow 0$ at infinity and zero structure constant $f^{abc} = 0$. But for non-Abelian theory, with larger enough A in low energy confinement case, it's possible for a non-zero α satisfying

$$\nabla^2 \alpha^a = g f^{abc} \vec{A}^c \cdot \vec{\nabla} \alpha^b. \quad (3.148)$$

But at least we solved the divergent problem due to infinite gauge choice with the price of introducing a delta function and an FP determinant. These two, however also make trouble because we cannot expand perturbation series directly. The latter is limited by the delta function that only valid around $G = 0$. Therefore, to compute FP determinant, we can do the infinitesimal transformation around $G = 0$. We first consider $G(A^u)$ as function of group element $u(\alpha)$, then change the integral measure in (3.132) to: $\mathcal{D}u = \mathcal{D}G \det \left| \frac{\delta \alpha}{\delta G} \right|$, so we have:

$$\Delta_G^{-1}[A] = \int \mathcal{D}G \det \left| \frac{\delta \alpha}{\delta G} \right| \delta(G). \quad (3.149)$$

Integration over gauge group G gives the FP determinant, then we get:

$$\Delta_G[A] = \det \left| \frac{\delta G}{\delta \alpha} \right|_{G=0}. \quad (3.150)$$

So (3.143) becomes:

$$\int \mathcal{D}A \delta(G(A)) \det \left| \frac{\delta G}{\delta \alpha} \right| e^{-S[A]}. \quad (3.151)$$

Next step we are going to put $\delta(G)$ and FP determinant into the exponential to get a pure path integral form. For that, we consider the Gauss integral

$$\int_{-\infty}^{+\infty} dc e^{-\frac{c^2}{2\xi}} \delta(G - c) = e^{-\frac{G^2}{2\xi}}. \quad (3.152)$$

Generating into the functional space, with $a = 1, \dots, d(G)$, and change the gauge $G^a = 0$ to $G^a = C^a$ with constants C^a . We have:

$$\int \mathcal{D}C^a e^{-\frac{1}{2\xi} \int d^4x C^a C^a} \delta(G^a - C^a) = e^{-\frac{1}{2\xi} \int d^4x G^a G^a}, \quad (3.153)$$

where we successfully put the delta function into the exponential. The price is adding a **gauge fixing term** in Lagrangian:

$$S_{\text{gf}} = \int d^4x \frac{1}{2\xi} G^a \cdot G^a. \quad (3.154)$$

Now we left a determinant needs to be eliminated: $\det \left| \frac{\delta G}{\delta \alpha} \right|$. This can be done by introducing the Grassmann number in the path integral:

$$\det \left| \frac{\delta G^a(\alpha(x))}{\delta \alpha^b(y)} \right| = \int \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{+ \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y)}, \quad (3.155)$$

where these two new field η and $\bar{\eta}$ is the **Faddeev-Popov Ghost fields**:

$$S_{\text{ghost}} = - \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.156)$$

Now we are ready to write the final form of Faddeev-Popov quantization:

$$\int \mathcal{D}A \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{-S[A, \bar{\eta}, \eta]}, \quad (3.157)$$

by inserting (3.154) and (3.155) into the Lagrangian:

$$S[A, \bar{\eta}, \eta] = S_{\text{YM}}[A] + S_{\text{gf}}[A] + S_{\text{ghost}}[A, \bar{\eta}, \eta] \quad (3.158)$$

$$= \int d^4x \left[\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} - \bar{\psi}(i\not{D} - m)\psi \right] + \frac{1}{2\xi} \int d^4x G^a(x) G^a(x) - \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.159)$$

It is worthy to note that we used **Euclidean convention**, which is

$$\int \mathcal{D}A e^{-S}. \quad (3.160)$$

If we use **Minkowski convention** with the sign $(+ - - -)$, which is

$$\int \mathcal{D}A e^{iS}. \quad (3.161)$$

So the action becomes

$$S_M[A, \bar{\eta}, \eta] = \int d^4x \left[\bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} \right] - \frac{1}{2\xi} \int d^4x G^a(x)G^a(x) + \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.162)$$

2.3 Feynman rules for non-Abelian gauge theory

The gauge condition we choose is the **covariant gauge (Lorenz gauge)**

$$G^a = \partial^\mu A_\mu^a = 0. \quad (3.163)$$

The quadratic terms in A_μ from $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c$ and $G^a = \partial^\mu A_\mu^a$ can be rewritten by the partial integration as:

$$S_{\text{quadratic}} = \int d^4x \left(\frac{1}{4}(\partial^\mu A_\nu^a - \partial^\nu A_\mu^a)(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a) + \frac{1}{2\xi}(\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a) \right) \quad (3.164)$$

$$= \int d^4x \left(\frac{1}{2}\partial^\mu A^{\nu a} \partial_\mu A_\nu^a - \frac{1}{2}\partial^\nu A^{\mu a} \partial_\mu A_\nu^a + \frac{1}{2\xi}(\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a) \right) \quad (3.165)$$

$$= \frac{1}{2} \int d^4x A_\mu^a \left(-\partial_\rho \partial^\rho g_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) \partial_\mu \partial_\nu \right) A^{\nu a} + \text{Boundary terms}. \quad (3.166)$$

Now with this gauge fixing term, we can check that the momentum operator becomes:

$$M_{\mu\nu} = -k^2 g_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) k_\mu k_\nu, \quad (3.167)$$

satisfying:

$$M_{\mu\nu} k^\nu = -k^2 k_\mu + \left(1 - \frac{1}{\xi}\right) k_\mu k^2 = -\frac{1}{\xi} k^2 k_\mu \neq 0. \quad (3.168)$$

Thus, we can then define the **propagators** legally by take inverse of momentum operator, which can be read from (3.166) as

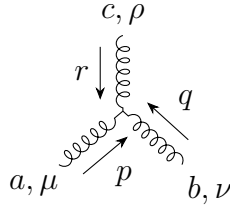
$$a, \mu \xrightarrow{p} b, \nu = \frac{-i\delta^{ab}}{p^2 + i\varepsilon} \left(g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right). \quad (3.169)$$

And the **gauge boson self-interaction** is vertices is given by the non-Abelian term in $\frac{1}{4}F^2$:

$$\mathcal{L}_{3A} = -gf^{abc}(\partial_\mu A_\nu^a)A^{\mu b}A^{\nu c} \quad (3.170)$$

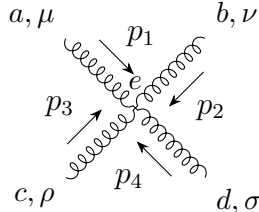
$$\mathcal{L}_{4A} = \frac{1}{4}g^2 f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \quad (3.171)$$

The first three gauge boson self interaction gives:



$$= -igf^{abc} [(r_\mu - q_\mu)g_{\nu\rho} + (q_\rho - p_\rho)g_{\mu\nu} + (p_\nu - r_\nu)g_{\mu\rho}] \quad (3.172)$$

The second four gauge boson self interaction gives:



$$= -ig^2 \left[f^{abe} f^{cde} (g_{\mu\rho} g_{\nu\sigma} - g_{\mu\sigma} g_{\nu\rho}) + f^{ace} f^{bde} (g_{\mu\nu} g_{\rho\sigma} - g_{\mu\sigma} g_{\nu\rho}) + f^{ade} f^{bce} (g_{\mu\nu} g_{\rho\sigma} - g_{\mu\rho} g_{\nu\sigma}) \right]. \quad (3.173)$$

Since we have introduced a new field: ghost field $\bar{\eta}, \eta$, we then can also find its Feynman rule from the Lagrangian. For this, we need to first calculate the derivative by (3.126) and the gauge $G^a = \partial^\mu A_\mu^a$:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{\delta G^a(x)}{\delta A_\mu^c(x)} \frac{\delta A_\mu^c(x)}{\delta \alpha^b(y)} = \frac{1}{g} \frac{\delta G^a(x)}{\delta A_\mu^c(x)} D_\mu^{bc} \delta(x-y) = \frac{1}{g} \partial^\mu D_\mu^{bc} \delta(x-y), \quad (3.174)$$

with the form of covariant derivative, we have:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \delta(x-y). \quad (3.175)$$

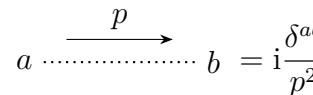
Thus the ghost action becomes:

$$S_{\text{ghost}} = \int d^4x \bar{\eta}^a(x) \left[\frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \right] \eta^b(x) \quad (3.176)$$

$$= \frac{1}{g} \int d^4x \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - \int d^4x f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \quad (3.177)$$

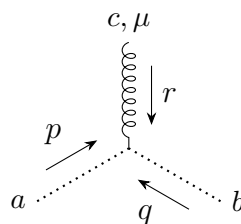
$$= \int d^4x \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - g \int d^4x f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \quad (3.178)$$

where we rescaled the field as $\eta \rightarrow \sqrt{g}\eta$ and $\bar{\eta} \rightarrow \sqrt{g}\bar{\eta}$ in the last step. Thus the first term is kinetic term and second is interaction term. So the kinetic term gives the **ghost propagators**:



$$a \cdots \cdots b = i \frac{\delta^{ab}}{p^2}. \quad (3.179)$$

And second interaction term gives the **vertex of ghost-ghost-gluon**:



$$= \frac{g}{2} f^{abc} (r^\mu + p^\mu - q^\mu) = -g f^{abc} q^\mu, \quad (3.180)$$

where the vertex energy-momentum conservation requires $r^\mu + p^\mu + q^\mu = 0$.

3. BRST Symmetry

3.1 Problem of negative norm

In last section, we introduced the gauge fixing term in the action to absorbing the gauge condition $G^a = 0$:

$$S_{\text{gf}} = \int d^4x \frac{1}{2\xi} G^a \cdot G^a, \quad (3.181)$$

which leads to the dynamical term of the gauge field as

$$\frac{1}{2\xi} (\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a). \quad (3.182)$$

However, this operation breaks the local gauge symmetry because it contains $\partial_0 A_0$. The problem of this term is A_0 has a negative norm:

$$\langle 0 | A_\mu(k) A_\nu^\dagger(k') | 0 \rangle \propto -\eta_{\mu\nu} \delta^{(4)}(k - k'), \quad (3.183)$$

thus

$$\langle 0 | A_0(k) A_0^\dagger(k') | 0 \rangle \propto -\delta^{(4)}(k - k') < 0. \quad (3.184)$$

This should not be a problem if local gauge symmetry was not broken, because there wouldn't be the term like $\partial_0 A_0$ get into the kinematic term, i.e., propagator. So the negative norm wasn't shown when we are doing real calculations. Now everything is on the table, the optical theorem tell us we need to count all the contributions including the negative norm final states. So it may cause the sum over all physical final states (positive norm states) is greater than 1. And since the gauge choice can be changed, then the value of the probability may no longer conserved or even depends on the gauge choice, which is unacceptable.

3.2 Infinitesimal BRST transformation

To solve this problem, Becchi, Rouet, Stora and Tyutin(BRST) found that there exist a global symmetry, namely the **BRST symmetry**, which allows us to separate the non-physical states.

To introduce this symmetry, we first consider the Gaussian integral formula:

$$\int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 \right\} = \sqrt{\frac{2\pi}{a}}. \quad (3.185)$$

Change the variable $B \rightarrow B + \frac{J}{a}$:

$$\int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} \left(B + \frac{J}{a} \right)^2 \right\} = \int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 - JB - \frac{J^2}{2a} \right\} = \text{Constant}. \quad (3.186)$$

Thus, dividing $\exp\left(\frac{J^2}{2a}\right)$ on both sides, we have

$$\exp \left\{ \frac{J^2}{2a} \right\} \propto \int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 - JB \right\}. \quad (3.187)$$

Now we can rewrite the gauge fixing term from the Faddeev-Popov Lagrangian by letting $J = \partial^\mu A_\mu^a$ and $a = -\xi$ as

$$\exp \left\{ i \int d^4x \left(-\frac{1}{2\xi} (\partial^\mu A_\mu^a)^2 \right) \right\} \propto \int \mathcal{D}B \exp \left\{ i \int d^4x \left(\frac{\xi}{2} (B^a)^2 + B^a (\partial^\mu A_\mu^a) \right) \right\}, \quad (3.188)$$

where B^a is an auxiliary field, which means this field is **non-dynamical**, because it doesn't contribute any term including $\partial_\mu B^a$. So the gauge fixing Lagrangian becomes:

$$\mathcal{L}_{\text{gf}} = \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A_\mu^a). \quad (3.189)$$

We can check this Lagrangian gives the same form with the Gaussian integration by implying the Euler-Lagrange equation:

$$\frac{\delta \mathcal{L}_{\text{gauge fix}}}{\delta B^a} = \xi B^a + \partial^\mu A_\mu^a = 0, \quad (3.190)$$

$$\implies B^a = -\frac{1}{\xi} \partial^\mu A_\mu^a. \quad (3.191)$$

So the Faddeev-Popov Lagrangian is

$$\mathcal{L}_{\text{FP}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} + \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A_\mu^a) + \bar{c}^a(-\partial^\mu D_\mu^{ab})c^b. \quad (3.192)$$

We have already derived variation of A from (3.126):

$$\delta A_\mu^a = D_\mu^{ab} \alpha^b(x), \quad (3.193)$$

where we absorbed the factor $1/g$ into the arbitrary function $\alpha^b(x) \rightarrow g\alpha^b(x)$. But we now want $\alpha^b(x)$ can related to the ghost field:

$$\alpha^b(x) = g\varepsilon c^b(x), \quad (3.194)$$

where ε is a constant that non-related with the space-time. So if we also do the same thing for the other fields' infinitesimal transformations, how will the Lagrangian change? This leads us to the **BRST transformations**. BRST claimed that there is a global symmetry from ε that the transformation keep the Lagrangian invariant

$$\delta_\varepsilon \mathcal{L} = 0. \quad (3.195)$$

And the infinitesimal parameter must be Grassmann-valued:

$$\varepsilon^2 = 0, \quad (3.196)$$

because A_μ^a is a bosonic field and c^b is a fermionic field, so ε must be a fermionic field, which fulfills the Grassmann algebra.

$$\varepsilon c^b = -c^b \varepsilon. \quad (3.197)$$

Thus the BRST transformation for A_μ^a and ψ_j can be derived by implying (3.194) into the normal infinitesimal transformation $\delta A_\mu^a = D_\mu^{ab} \alpha^b(x)$ and $\delta_\varepsilon \psi_j = i \sum_a \alpha^a (t^a \psi)_j$:

$$\delta_\varepsilon A_\mu^a = \varepsilon D_\mu^{ab} c^b = \varepsilon \left(\partial_\mu c^a + g \sum_{b,c} f^{abc} A_\mu^b c^c \right), \quad (3.198)$$

$$\delta_\varepsilon \psi_j = ig\varepsilon \sum_a c^a (t^a \psi)_j. \quad (3.199)$$

These two transformations obviously keep the $\mathcal{L}_{\text{YM}}$ invariant, because it just changes a name of α , and since αc^a is still a local gauge transformation, then $\mathcal{L}_{\text{YM}}$ is automatically invariant under this transformation. One can check that $F_{\mu\nu}^a$ follows

$$\delta_\varepsilon F_{\mu\nu}^a = g\varepsilon f^{abc} F_{\mu\nu}^b c^c, \quad (3.200)$$

thus

$$\delta_\varepsilon \left(-\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} \right) = 0. \quad (3.201)$$

And for the fermionic field, the covariant derivative follows:

$$\delta_\varepsilon (D_\mu \psi) = ig\varepsilon c^a t^a (\partial_\mu \psi - igA_\mu^b t^b \psi) = ig\varepsilon c^a t^a (D_\mu \psi), \quad (3.202)$$

thus

$$\delta_\varepsilon (\bar{\psi} (i\gamma^\mu D_\mu - m) \psi) = (\delta_\varepsilon \bar{\psi}) (i\gamma^\mu D_\mu - m) \psi + \bar{\psi} (i\gamma^\mu \delta_\varepsilon (D_\mu \psi) - m \delta_\varepsilon \psi) = 0. \quad (3.203)$$

But it's worthy to mention that **BRST transformation is not a local gauge transformation (transformation parameter ε is global)**. But it indeed **comes from a local gauge transformation (transformation parameter $\alpha^a(x) = g\varepsilon c^a(x)$ is local)**.

Now it comes to the most tedious part, the gauge fixing term and the ghost term. This is because these two terms are not gauge invariant. For convenient, we can just check the following variation of the ghost field and auxiliary field:

$$\delta_\varepsilon c^a = -\frac{1}{2} g\varepsilon \sum_{b,c} f^{abc} c^b c^c \quad (3.204)$$

$$\delta_\varepsilon \bar{c}^a = \varepsilon B^a \quad (3.205)$$

$$\delta_\varepsilon B^a = 0, \quad (3.206)$$

keep the Lagrangian of gauge fixing and ghost invariant:

$$\delta_\varepsilon (\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}) = 0, \quad (3.207)$$

namely to prove:

$$\delta_\varepsilon \left(\frac{\xi}{2} (B^a)^2 \right) + \underbrace{(\delta_\varepsilon B^a)}_{=0} (\partial^\mu A_\mu^a) + B^a \underbrace{\delta_\varepsilon (\partial^\mu A_\mu^a)}_{=\partial^\mu (\varepsilon D_\mu^{ab} c^b)} - \underbrace{(\delta_\varepsilon \bar{c}^a)}_{\varepsilon B^a} (\partial^\mu D_\mu^{ab}) c^b + \underbrace{\bar{c}^a \delta_\varepsilon (\partial^\mu D_\mu^{ab}) c^b + \bar{c}^a (\partial^\mu D_\mu^{ab}) (\delta_\varepsilon c^b)}_{\bar{c}^a \partial^\mu \delta_\varepsilon (D_\mu^{ab} c^b)} = 0. \quad (3.208)$$

The first term is simply 0 because

$$\delta_\varepsilon (B^a)^2 = 2\delta_\varepsilon B^a = 0 \quad (3.209)$$

And for the third term

$$\partial^\mu (\varepsilon D_\mu^{ab} c^b) = \underbrace{(\partial^\mu \varepsilon)}_{=0} D_\mu^{ab} c^b + \varepsilon \partial^\mu (D_\mu^{ab} c^b) = \varepsilon \partial^\mu (D_\mu^{ab} c^b), \quad (3.210)$$

due to ε is a constant. Now the expression to be proved reduces to

$$\cancel{B^a \varepsilon \partial^\mu D_\mu^{ab} c^b} - \varepsilon B^a \partial^\mu D_\mu^{ab} c^b - \bar{c}^a \partial^\mu (\delta_\varepsilon D_\mu^{ab} c^b) \quad (3.211)$$

$$= -\bar{c}^a \partial^\mu \left\{ \partial_\mu (\delta_\varepsilon c^a) + g \sum_{b,c} f^{abc} (\delta_\varepsilon A_\mu^b) c^c + g \sum_{b,c} f^{abc} A_\mu^b (\delta_\varepsilon c^c) \right\} \quad (3.212)$$

$$= -\bar{c}^a \partial^\mu \left\{ \partial_\mu \left(-\frac{g}{2} \varepsilon \sum_{b,c} f^{abc} c^b c^c \right) + g \sum_{b,c} f^{abc} \left[\varepsilon \left(\partial_\mu c^b + g \sum_{d,e} f^{bde} A_\mu^d c^e \right) \right] c^c \right. \\ \left. + g \sum_{b,c} f^{abc} A_\mu^b \left(-\frac{g}{2} \varepsilon \sum_{d,e} f^{cde} c^d c^e \right) \right\} \quad (3.213)$$

$$= -\bar{c}^a \partial^\mu \left\{ g \varepsilon \sum_{b,c} f^{abc} \left[-\frac{1}{2} (\partial_\mu c^b c^c) + (\partial_\mu c^b) c^c \right] + g^2 \varepsilon \sum_{b,c,e,f} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right) \right\} \quad (3.214)$$

The first term will be canceled by using the property of Grassmann number $c^a c^b = -c^b c^a$ and structure constant $f^{abc} = -f^{acb}$:

$$\sum_{b,c} f^{abc} \partial_\mu (c^b c^c) = \sum_{b,c} f^{abc} [(\partial_\mu c^b) c^c + c^b (\partial_\mu c^c)] \quad (3.215)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c + f^{acb} c^c (\partial_\mu c^b)] \quad (3.216)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c - f^{abc} c^c (\partial_\mu c^b)] \quad (3.217)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c + f^{abc} (\partial_\mu c^b) c^c] \quad (3.218)$$

$$= 2 \sum_{b,c} f^{abc} (\partial_\mu c^b) c^c. \quad (3.219)$$

Now the only left term is

$$-\bar{c}^a \partial^\mu g^2 \varepsilon \sum_{b,c,e,f} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right), \quad (3.220)$$

which can be canceled by Jacobi identity

$$f^{ade} f^{bcd} + f^{bde} f^{cad} + f^{cde} f^{abd} = 0. \quad (3.221)$$

For that, we first relabel the indices d, e in the first term:

$$\sum_{b,c,d,e} f^{abc} f^{bde} c^e c^c = \frac{1}{2} \sum_{b,c,d,e} (f^{abc} f^{bde} - f^{abe} f^{bdc}) c^e c^c. \quad (3.222)$$

Thus,

$$\sum_{b,c,d,e} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right) \quad (3.223)$$

$$= \sum_{b,c,d,e} f^{abc} f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{abc} f^{cde} A_\mu^b c^d c^e \quad (3.224)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{abc} f^{bde} - f^{abe} f^{bdc}) A_\mu^d c^e c^c - f^{abc} f^{cde} A_\mu^b c^d c^e \quad (3.225)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{ace} f^{cbd} - f^{acd} f^{cbe} - f^{abc} f^{cde}) A_\mu^b c^d c^e \quad (3.226)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (-f^{dbc} f^{ace} + f^{ebc} f^{acd} - f^{abc} f^{cde}) A_\mu^b c^d c^e \quad (3.227)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{dbc} f^{aec} - f^{ebc} f^{adc} - f^{abc} f^{dec}) A_\mu^b c^d c^e \quad (3.228)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{dbc} f^{aec} + f^{ebc} f^{dac} + f^{abc} f^{edc}) A_\mu^b c^d c^e \quad (3.229)$$

$$= 0. \quad (3.230)$$

Finally, with such tedious calculation, we have successfully proved (3.207) being satisfied, which means the gauge fixing term and ghost term together are BRST invariant. Thus, the Faddeev-Popov Lagrangian is BRST invariant:

$$\delta_\varepsilon \mathcal{L}_{\text{FP}} = \delta_\varepsilon \mathcal{L}_{\text{YM}} + \delta_\varepsilon (\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}) = 0. \quad (3.231)$$

3.3 Factorization of Hilbert space via BRST operator

Now since BRST symmetry is a global symmetry, we can in principle define a conservation charge Q , which should upgrade to an operator in Hilbert space since we have already quantized the classical field. This operator thus called the **BRST operator** Q , which can be easily defined from the infinitesimal transformation:

$$\delta_\varepsilon \Phi = [\varepsilon Q, \Phi] = \varepsilon(Q\Phi); \quad \text{for bosonic field } \Phi \quad (3.232)$$

$$\delta_\varepsilon \Psi = \{\varepsilon Q, \Psi\} = -\varepsilon(Q\Psi); \quad \text{for fermionic field } \Psi. \quad (3.233)$$

Thus Q has similar properties with δ_ε , which can be directly copy from (3.198),(3.199),(3.204),(3.205) and (3.206):

$$QA_\mu^a = D_\mu^{ab} c^b \quad (3.234)$$

$$Q\psi_j = igc^a(t^a\psi_j) \quad (3.235)$$

$$Qc^a = -\frac{1}{2}gf^{abc}c^bc^c \quad (3.236)$$

$$Q\bar{c}^a = B^a \quad (3.237)$$

$$QB^a = 0, \quad (3.238)$$

and with (3.196), one can prove that Q satisfies **Nilpotency**:

$$Q^2 = 0, \quad (3.239)$$

by acting on the gauge field A_μ^a , the ghost and anti-ghost field c^a and $\bar{c}^a$ and the auxiliary field B^a respectively to show the results are all vanished.

With BRST operator, we can rewrite the Faddeev-Popov Lagrangian as

$$\mathcal{L}_{\text{FP}} = \mathcal{L}_{\text{YM}} + \underbrace{Q \left(\bar{c}^a \partial^\mu A_\mu^a + \frac{\xi}{2} \bar{c}^a B^a \right)}_{\text{BRST ancestors}}. \quad (3.240)$$

With this form, we can easily see that it's BRST invariant by using (3.239) and $\delta_\varepsilon \mathcal{L}_{\text{YM}} = 0$:

$$Q\mathcal{L}_{\text{FP}} = 0, \quad Q\mathcal{L}_{\text{YM}} = 0 \quad (3.241)$$

Now we can use BRST operator to show how those “nonphysical” states are separated. Since BRST operator as a conserved “charge”. It naturally commutes with Hamiltonian operator H :

$$[Q, H] = 0, \quad Q = Q^\dagger. \quad (3.242)$$

The Hilbert space of eigenstates of H in Heisenberg picture can be formally expressed as

$$\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_0, \quad (3.243)$$

where the subspace can be defined by the map:

$$Q : \mathcal{H} \rightarrow \mathcal{H}. \quad (3.244)$$

The kernel of Q , i.e., those states are mapped to 0, is

$$\text{Ker}(Q) = \{|\psi\rangle \in \mathcal{H} \mid Q|\psi\rangle = 0\}, \quad (3.245)$$

and the image is

$$\text{Im}(Q) = \{|\psi\rangle \in \mathcal{H} \mid |\psi\rangle = Q|\phi\rangle, \text{ for } |\phi\rangle \in \mathcal{H}, \}. \quad (3.246)$$

And since $Q^2 = 0$, so for any states in $\text{Im}(Q)$ are all in $\text{Ker}(Q)$ too, i.e.,

$$\text{Im}(Q) \subset \text{Ker}(Q). \quad (3.247)$$

Thus, we define the three states as follows:

$$\mathcal{H}_1 = \mathcal{H} \setminus \text{Ker}(Q) = \{|\psi_1\rangle \in \mathcal{H} \mid Q|\psi_1\rangle \neq 0\}, \quad (3.248)$$

$$\mathcal{H}_2 = \text{Im}(Q) = \{|\psi_2\rangle \in \mathcal{H} \mid |\psi_2\rangle = Q|\phi\rangle, |\phi\rangle \in \mathcal{H}_1\}, \quad (3.249)$$

$$\mathcal{H}_0 = \text{Ker}(Q) \setminus \text{Im}(Q) = \{|\psi_0\rangle \in \mathcal{H} \mid Q|\psi_0\rangle = 0 \text{ and } \forall |\phi\rangle \in \mathcal{H}, |\psi\rangle \neq Q|\phi\rangle\}. \quad (3.250)$$

With these definition, we can clearly see that Hilbert space is factorized into the direct sum over three subspace :

$$\mathcal{H} = \underbrace{(\mathcal{H} \setminus \text{Ker}(Q))}_{\mathcal{H}_1} \oplus \underbrace{\text{Im}(Q)}_{\mathcal{H}_2} \oplus \underbrace{(\text{Ker}(Q) \setminus \text{Im}(Q))}_{\mathcal{H}_0}. \quad (3.251)$$

With this, we claim the set of all physical states is the kernel of $\mathcal{H}$ mode the image:

$$\mathcal{H}_{\text{phys}} \cong \mathcal{H}_0 = \text{coh}(Q) = \frac{\text{Ker}(Q)}{\text{Im}(Q)}, \quad (3.252)$$

which is a **cohomology space** satisfying:

$$Q|\psi\rangle = 0. \quad (3.253)$$

To show that it indeed make sense, we need to go back to the free theory, namely $g \rightarrow 0$, so the BRST transformation reduce to

$$QA_\mu^a = \partial_\mu c^a, \quad (3.254)$$

$$Q\bar{c}^a = B^a, \quad (3.255)$$

$$Q\psi_j = Qc^a = QB^a = 0. \quad (3.256)$$

We find this result with the definition of $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_0$ can help us to classify these asymptotic states:

$$A_\mu^L \in \mathcal{H}_1, \quad \bar{c}^a \in \mathcal{H}_1; \quad (3.257)$$

$$A_\mu^T \in \mathcal{H}_0, \quad \psi_j \in \mathcal{H}_0; \quad (3.258)$$

$$c^a \in \mathcal{H}_2, \quad B^a \in \mathcal{H}_2. \quad (3.259)$$

where A_μ^L represent the longitudinal polarization

$$A_\mu^L \sim k_\mu \phi \implies Q|A_\mu^L\rangle \sim |c^a\rangle \neq 0, \quad (3.260)$$

and A_μ^T represent the transversal polarization

$$k^\mu A_\mu^T = 0 \implies QA_\mu^T \sim k_\mu c^a = 0. \quad (3.261)$$

With this classification, we can see that for the states in $\mathcal{H}_1$, like A_μ^L or $\bar{c}^a$, the norm of time component are **negative**:

$$\langle A_0^L | A_0^L \rangle = \langle 0 | a_0 a_0^\dagger | 0 \rangle = \langle 0 | [a_0, a_0^\dagger] | 0 \rangle = -1 < 0, \quad (3.262)$$

$$\langle \bar{\eta} | \bar{\eta} \rangle = \langle 0 | \bar{c} c^\dagger | 0 \rangle = \langle 0 | (-c^\dagger)^\dagger c^\dagger | 0 \rangle = -\langle 0 | \bar{c} c^\dagger | 0 \rangle = \langle 0 | \{c, c^\dagger\} | 0 \rangle = -1 < 0. \quad (3.263)$$

For the states in $\mathcal{H}_2$ are **zero norm** due to the nilpotency $Q^2 = 0$:

$$\langle \psi_2 | \psi_2 \rangle = \langle \psi_1 | Q^\dagger Q | \psi_1 \rangle = \langle \psi_1 | Q^2 | \psi_1 \rangle \equiv 0. \quad (3.264)$$

And the states in $\mathcal{H}_0$ are **positive norm**, which is the real physical state space. So we say, physical degrees of freedom are elements of $\mathcal{H}_0$, and unphysical degrees of freedom are in $\mathcal{H}_1, \mathcal{H}_2$.

This result however, one may wondering if it is picture dependent since we assume the Heisenberg picture that the states doesn't evolve with time. The answer is lucky no, picture does not change under time evolution: for any state $|\psi_0, t\rangle$ in $\mathcal{H}$, we have

$$Q|\psi_0, t\rangle = Qe^{iHt}|\psi_0\rangle = e^{iHt}Q|\psi_0\rangle = 0, \quad (3.265)$$

which means $|\psi_0, t\rangle$ must be a linear combination of states from $\mathcal{H}_0$ and $\mathcal{H}_2$, i.e., $|\psi_0, t\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2$. But if we consider a scattering process from $|\psi_0, t\rangle$ to any $|\psi_2\rangle \in \mathcal{H}_2$ that satisfies $|\psi_2\rangle = Q|\psi_1\rangle$, we have

$$\langle \psi_2 | \psi_0, t \rangle = \langle \psi_1 | Q | \psi_0, t \rangle = 0, \quad (3.266)$$

which means the scalar product vanishes! Namely, $|\psi_0, t\rangle$ has no effective component in $\mathcal{H}_2$. So even though we mix some element of $\mathcal{H}_2$ in the combination of the construction of $|\psi_0, t\rangle$, it doesn't matter because the elements in $\mathcal{H}_2$ won't contribute to the final result.

Thus, we find that **BRST operator factorize the space based on positive norm, negative norm and zero norm**. which helps us to select and drop the negative and zero norm states.

4. Renormalization of non-Abelian gauge theory

Now we have enough foundation to consider the renormalization of Padeev–Popov theory. For that, we first recall the Feynman rules of Faddeev–Popov Lagrangian for a Yang–Mills theory coupled to fermions,

$$\mathcal{L}_{\text{FP}} = \mathcal{L}_{\text{YM}} - \frac{1}{2\xi} (\partial^\mu A_\mu^a) (\partial^\nu A_\nu^a) + \bar{c}^a (-\partial^\mu D_\mu^{ab}) c^b. \quad (3.267)$$

It is useful to separate it into the free part and the interaction part:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}. \quad (3.268)$$

In Feynman gauge, $\xi = 1$, the free Lagrangian reads

$$\mathcal{L}_0 = \bar{\psi} (i\not{\partial} - m) \psi + \frac{1}{2} A_\mu^a (g^{\mu\nu} \square - \partial^\mu \partial^\nu) A_\nu^a. \quad (3.269)$$

The corresponding fermion and gauge boson propagators are

$$\begin{array}{c} p \\ \longrightarrow \end{array} = \frac{i}{\not{p} - m + i\varepsilon} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}, \quad (3.270)$$

$$a, \mu \begin{array}{c} \xrightarrow{p} \\ \text{~~~~~} \end{array} b, \nu = \frac{-i \delta^{ab}}{p^2 + i\varepsilon} \left[g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2 + i\varepsilon} \right]. \quad (3.271)$$

The interaction terms are

$$\begin{aligned} \mathcal{L}_{\text{int}} = & g A_\mu^a (\bar{\psi} \gamma^\mu t^a \psi) - g f^{abc} (\partial_\mu A_\nu^a) A^{b\mu} A^{c\nu} \\ & - \frac{1}{4} g^2 (f^{abc} A_\mu^b A_\nu^c) (f^{ade} A^{d\mu} A^{e\nu}) + g f^{abc} (\partial^\mu \bar{c}^a) A_\mu^b c^c. \end{aligned} \quad (3.272)$$

With all momenta conventionally taken incoming, the basic vertices are:

$$\begin{array}{c} \mu, a \\ \text{~~~~~} \\ \swarrow \quad \searrow \\ i \quad \quad j \end{array} = ig \gamma^\mu (t^a)_{ij}, \quad (3.273)$$

$$\begin{array}{c} \mu, a, p \\ \text{~~~~~} \\ \swarrow \quad \searrow \\ \nu, b, q \quad \rho, c, r \end{array} = g f^{abc} [g^{\mu\nu} (p - q)^\rho + g^{\nu\rho} (q - r)^\mu + g^{\rho\mu} (r - p)^\nu], \quad (3.274)$$

$$\begin{array}{c} \mu, a \quad \nu, b \\ \text{~~~~~} \quad \text{~~~~~} \\ \swarrow \quad \searrow \\ \rho, c \quad \sigma, d \end{array} = -ig^2 \left[f^{abe} f^{cde} (g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho}) \right. \\ \left. + f^{ace} f^{bde} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho}) \right. \\ \left. + f^{ade} f^{bce} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma}) \right]. \quad (3.275)$$

For the ghost propagator and ghost–gluon vertex one has

$$a \xrightarrow{p} b = i \frac{\delta^{ab}}{p^2}; \quad (3.276)$$

$$\begin{array}{c} c, \mu \\ \downarrow r \\ \begin{array}{ccc} & \text{---} & \\ p \nearrow & & \searrow q \\ a & & b \end{array} \end{array} = \frac{g}{2} f^{abc} (r^\mu + p^\mu - q^\mu) = -g f^{abc} q^\mu, \quad (3.277)$$

which are the result that we should be familiar with in previous section.

4.1 Why the Faddeev–Popov theory is renormalizable

Before introducing the explicit counterterms, we should explain why the Faddeev–Popov theory is expected to be renormalizable at all. The point is not to guess the counterterms first, but to prove that the ultraviolet divergent part of the theory can only have the same local form as the original Faddeev–Popov Lagrangian. This statement follows from two ingredients: **power counting** and **BRST symmetry**.

Limitation from power counting

We work in four space-time dimensions. Just as we did in Chapter 2. Since the action is dimensionless,

$$[S] = 0, \quad [d^4x] = -4, \quad [\mathcal{L}] = 4. \quad (3.278)$$

The dimensions of the fields are fixed by the quadratic terms. From the gauge kinetic term,

$$-\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} \supset -\frac{1}{2} (\partial_\mu A_\nu^a) (\partial^\mu A^{a\nu}), \quad (3.279)$$

we get

$$2(1 + [A_\mu]) = 4 \quad \implies \quad [A_\mu] = 1. \quad (3.280)$$

From the fermion kinetic term,

$$\bar{\psi} i \gamma^\mu \partial_\mu \psi, \quad (3.281)$$

we get

$$[\bar{\psi}] + 1 + [\psi] = 4, \quad [\bar{\psi}] = [\psi], \quad \implies \quad [\psi] = [\bar{\psi}] = \frac{3}{2}. \quad (3.282)$$

Finally, from the ghost kinetic term,

$$-\bar{c}^a \partial_\mu \partial^\mu c^a, \quad (3.283)$$

we get

$$[\bar{c}] + 2 + [c] = 4. \quad (3.284)$$

The natural symmetric assignment is

$$[c] = [\bar{c}] = 1. \quad (3.285)$$

Therefore the dimensions of all interaction terms are

$$[A_\mu \bar{\psi} \gamma^\mu \psi] = 1 + \frac{3}{2} + \frac{3}{2} = 4, \quad (3.286)$$

$$[(\partial A)AA] = 1 + 1 + 1 + 1 = 4, \quad (3.287)$$

$$[AAAA] = 4, \quad (3.288)$$

$$[(\partial \bar{c})Ac] = 1 + 1 + 1 + 1 = 4. \quad (3.289)$$

Thus the coupling g is dimensionless:

$$[g] = 0. \quad (3.290)$$

This is the first indication of renormalizability just like (2.10). A negative-dimensional coupling would force higher and higher powers of momentum to appear at higher orders, producing infinitely many independent divergent structures. Here this does not happen.

Now we make the statement precise by computing the superficial degree of divergence. At large loop momentum, the propagators in (3.270), (3.271) and (3.276), behave as

$$S_\psi(k) \sim \frac{\not{k}}{k^2} \sim \frac{1}{k}, \quad D_A(k) \sim \frac{1}{k^2}, \quad D_c(k) \sim \frac{1}{k^2}. \quad (3.291)$$

The three-gluon vertex and the ghost-gluon vertex each contain one derivative, which in momentum space becomes a off-shell momentum that need to be integrated, so they contribute one power of loop momentum in the numerator. The fermion-gluon vertex and the four-gluon vertex contain no derivative.

Let

$$I_A, I_c, I_\psi \quad (3.292)$$

be the numbers of internal gauge-boson, ghost and fermion lines. Let

$$V_3, V_4, V_c, V_\psi \quad (3.293)$$

denote the numbers of AAA , $AAAA$, $A\bar{c}c$ and $A\bar{\psi}\psi$ vertices. Let the corresponding external lines be

$$E_A, E_c, E_\psi, \quad (3.294)$$

where E_c counts external ghost and anti-ghost lines together, and E_ψ counts external fermion and anti-fermion lines together. The superficial degree of divergence in $d = 4$ from (2.1) is then

$$D = 4L - 2I_A - 2I_c - I_\psi + V_3 + V_c. \quad (3.295)$$

The terms in this formula have the following origin:

- each **loop integral** contributes $+4$ (integral measure d^4k);
- each **internal gauge** or **ghost propagator** contributes -2 ($\sim \frac{1}{k^2}$);
- each **internal fermion propagator** contributes -1 ($\sim \frac{\not{k}}{k^2}$);
- each **gauge boson vertex** $(\partial A)AA$ or **ghost-boson vertex** $\partial \bar{c}Ac$ vertex contributes $+1$ from its derivative ($i\partial \rightarrow k$).

The topological relation between loops, internal lines and vertices is

$$L = I_A + I_c + I_\psi - (V_3 + V_4 + V_c + V_\psi) + 1. \quad (3.296)$$

We also have the line-counting identities

$$2I_\psi + E_\psi = 2V_\psi, \quad (3.297)$$

$$2I_c + E_c = 2V_c, \quad (3.298)$$

$$2I_A + E_A = 3V_3 + 4V_4 + V_c + V_\psi. \quad (3.299)$$

Substituting the loop relation into D gives

$$\begin{aligned} D &= 4(I_A + I_c + I_\psi - V_3 - V_4 - V_c - V_\psi + 1) - 2I_A - 2I_c - I_\psi + V_3 + V_c \\ &= 4 + 2I_A + 2I_c + 3I_\psi - 3V_3 - 4V_4 - 3V_c - 4V_\psi. \end{aligned} \quad (3.300)$$

Now use the three line-counting identities:

$$2I_A = 3V_3 + 4V_4 + V_c + V_\psi - E_A, \quad (3.301)$$

$$2I_c = 2V_c - E_c, \quad (3.302)$$

$$3I_\psi = 3V_\psi - \frac{3}{2}E_\psi. \quad (3.303)$$

Hence

$$\begin{aligned} D &= 4 + (3V_3 + 4V_4 + V_c + V_\psi - E_A) + (2V_c - E_c) + \left(3V_\psi - \frac{3}{2}E_\psi\right) \\ &\quad - 3V_3 - 4V_4 - 3V_c - 4V_\psi \\ &= 4 - E_A - E_c - \frac{3}{2}E_\psi. \end{aligned} \quad (3.304)$$

This is the crucial power-counting result:

$$\boxed{D = 4 - E_A - E_c - \frac{3}{2}E_\psi.} \quad (3.305)$$

The important point is that D is independent of the number of vertices. Thus **going to higher loop order does not create more and more divergent structures**. Only Green functions with sufficiently few external legs can be superficially divergent. For example,

- Two gauge boson external lines (gauge 1PI):

$$AA : \quad D = 2. \quad (3.306)$$

- Two ghost external lines (ghost 1PI):

$$\bar{c}c : \quad D = 2. \quad (3.307)$$

- Two fermion external lines (fermion 1PI):

$$\bar{\psi}\psi : \quad D = 1. \quad (3.308)$$

- Three gauge external lines:

$$AAA : \quad D = 1. \quad (3.309)$$

- Three gauge-ghost external lines:

$$A\bar{c}c : \quad D = 1. \quad (3.310)$$

- Three gauge-fermion external lines:

$$A\bar{\psi}\psi : \quad D = 0. \quad (3.311)$$

- Four gauge external lines:

$$AAAA : \quad D = 0. \quad (3.312)$$

For amplitudes with more external fields, $D < 0$, which is (superficially) convergent after subdivergences have been subtracted, namely we don't need to consider the renormalization of terms like $AAAAA$. So we only need to consider the divergent in $D \leq 4$

Limitation from BRST symmetry: Slavnov-Taylor identity

However, power counting alone is not yet the full proof. It tells us that the divergent terms are local and have dimension no larger than 4, but it does not by itself explain why these local terms must combine into the gauge-invariant Faddeev–Popov structure. For this we use BRST symmetry. It is most transparent to introduce the auxiliary field B^a and write

$$\mathcal{L}_{\text{FP}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \bar{\psi}(i\not{D} - m)\psi + \frac{\xi}{2}(B^a)^2 + B^a \partial^\mu A_\mu^a + \bar{c}^a (-\partial^\mu D_\mu^{ab})c^b. \quad (3.313)$$

The BRST transformations are

$$QA_\mu^a = D_\mu^{ab}c^b, \quad (3.314)$$

$$Qc^a = -\frac{g}{2}f^{abc}c^b c^c, \quad (3.315)$$

$$Q\bar{c}^a = B^a, \quad (3.316)$$

$$QB^a = 0, \quad (3.317)$$

$$Q\psi = igc^a t^a \psi, \quad (3.318)$$

$$Q\bar{\psi} = -ig\bar{\psi}c^a t^a. \quad (3.319)$$

As shown in the BRST section of the notes, these transformations are nilpotent,

$$Q^2 = 0, \quad (3.320)$$

and the Faddeev–Popov Lagrangian is BRST invariant:

$$Q\mathcal{L}_{\text{FP}} = 0. \quad (3.321)$$

Next, for understanding how the BRST symmetry limits the renormalization counter terms, we need to briefly introduce the Slavnov–Taylor identity. The idea is not mysterious: it is simply the Ward identity associated with BRST symmetry. In the path integral we integrate over all fields as

$$Z[J] = \int \mathcal{D}\Phi \exp \left\{ iS_{\text{FP}}[\Phi] + i \int J\Phi \right\}, \quad (3.322)$$

where Φ denotes all fields, for example $A_\mu, c, \bar{c}, B, \psi, \bar{\psi}$. And S_{FP} is the **classical action**, by contrast the **effective action** can be expressed as:

$$\Gamma[\phi_{\text{cl}}] = W[J] - \int d^4x J(x)\phi_{\text{cl}}(x), \quad (3.323)$$

where $W[J] = -i \ln Z[J]$, and $\phi_{\text{cl}}(x) = \frac{\delta W[J]}{\delta J(x)} = \langle \phi(x) \rangle_J$. It can be related with the source by

$$\frac{\delta \Gamma[\phi_{\text{cl}}]}{\delta \phi_{\text{cl}}(x)} = -J(x). \quad (3.324)$$

Thus, the effective action $\Gamma[\phi_{\text{cl}}]$ gives the quantum corrected equation of motion instead of the classical one given by $S[\phi]$ by taking $J = 0$. If we make the BRST change of integration variables

$$\Phi \longrightarrow \Phi + \varepsilon Q\Phi, \quad (3.325)$$

with ε an infinitesimal Grassmann parameter, the value of the path integral cannot change. The reason is the same as in ordinary calculus: changing integration variables does not change the integral. If the action is BRST invariant and the regularization also preserves this symmetry, then both the measure and S_{FP} are unchanged. Therefore the only variation comes from the source term. This gives a Ward identity for the generating functional. After performing the Legendre transform from the connected generating functional to the quantum effective action Γ , this Ward identity is written as an equation of effective action Γ :

$$\mathcal{S}(\Gamma) = 0. \quad (3.326)$$

This equation is called the **Slavnov–Taylor identity**. Thus, in simple words, the Slavnov–Taylor identity says: **the full quantum effective action must still respect the BRST version of gauge symmetry**. Here Γ is important because it generates one-particle-irreducible amplitudes. Therefore $\mathcal{S}(\Gamma) = 0$ is not just a formal statement; it relates the possible divergent parts of propagators and vertices.

Next we explain why this constrains counterterms. Expand the effective action in loops:

$$\Gamma = S_{\text{FP}} + \hbar \Gamma^{(1)} + \hbar^2 \Gamma^{(2)} + \dots \quad (3.327)$$

At a fixed loop order, say at n loops, the contribution can be separated into a divergent part and a finite part:

$$\Gamma^{(n)} = \Gamma_{\text{div}}^{(n)} + \Gamma_{\text{fin}}^{(n)}. \quad (3.328)$$

In dimensional regularization the divergent part has the schematic local form

$$\Gamma_{\text{div}}^{(n)} = \sum_i \frac{a_i^{(n)}}{\epsilon} \int d^4x \mathcal{O}_i(x), \quad d = 4 - \epsilon. \quad (3.329)$$

The symbols $\mathcal{O}_i$ denote local operators built from the fields and their derivatives. For example, possible operators one might first write down are

$$A_\mu^a A^{a\mu}, \quad F_{\mu\nu}^a F^{a\mu\nu}, \quad \bar{\psi} i \not{D} \psi. \quad (3.330)$$

Only some of them will actually be allowed after imposing BRST symmetry.

From now on, when we analyze one fixed loop order, we use the shorter notation

$$\Delta \equiv \Gamma_{\text{div}}^{(n)} \quad (3.331)$$

with the overall factor $\hbar^n$ suppressed. Thus Δ is not a new interaction invented by hand. It is the possible ultraviolet divergent part produced by loop integrals. Equivalently,

$$\Delta = \int d^4x \mathcal{L}_{\text{div}}^{(n)}. \quad (3.332)$$

The counterterm action is introduced only after this divergence has appeared. At the same loop order we add

$$S_{\text{ct}}^{(n)} = -\Gamma_{\text{div}}^{(n)} = -\Delta, \quad (3.333)$$

up to possible finite scheme-dependent counterterms. Then the divergent part of the effective action is cancelled:

$$\Gamma_{\text{div}}^{(n)} + S_{\text{ct}}^{(n)} = \Delta - \Delta = 0. \quad (3.334)$$

This is the precise relation between Δ and the counterterm. Strictly speaking, Δ denotes the divergent local functional, while the counterterm added to the action is $-\Delta$. In common language one often calls the local structure Δ itself a “candidate counterterm”, because the same operator appears in the counterterm action with the opposite coefficient. Therefore, asking whether a candidate Δ is allowed means asking whether a loop divergence of that local form can appear and, if it appears, whether it can be cancelled by a local counterterm. First introduce external sources, usually called BRST sources or antifields, which record the BRST variations of the fields. For the present discussion we write the extended classical action as

$$S_{\text{ext}} = S_{\text{FP}} + \int d^4x \left[K_A^{a\mu} Q A_\mu^a + K_c^a Q c^a + K_\psi Q \psi + Q \bar{\psi} K_{\bar{\psi}} \right]. \quad (3.335)$$

The K ’s are not new physical fields. They are only a bookkeeping device. At the classical level their job is to make the following identities true:

$$\frac{\delta S_{\text{ext}}}{\delta K_A^{a\mu}} = Q A_\mu^a, \quad \frac{\delta S_{\text{ext}}}{\delta K_c^a} = Q c^a, \quad \frac{\delta S_{\text{ext}}}{\delta K_\psi} = Q \psi, \quad \frac{\delta S_{\text{ext}}}{\delta K_{\bar{\psi}}} = Q \bar{\psi}. \quad (3.336)$$

Now define the path integral with two kinds of sources:

$$Z[J, K] = \int \mathcal{D}\Phi \exp \left\{ i S_{\text{ext}}[\Phi, K] + i \int d^4x J_\Phi \Phi \right\}. \quad (3.337)$$

Here J_Φ is the ordinary source coupled to the field Φ , while K_Φ is the source coupled to the composite BRST variation $Q\Phi$. Define

$$W[J, K] = -i \ln Z[J, K]. \quad (3.338)$$

Because K_A is coupled to QA , differentiating W with respect to K_A inserts QA into the path integral:

$$\frac{\delta W}{\delta K_A^{a\mu}(x)} = \langle Q A_\mu^a(x) \rangle_{J, K}, \quad (3.339)$$

and similarly for $K_c, K_\psi, K_{\bar{\psi}}$. Then we can pass from W to the effective action. Define classical fields by

$$\Phi(x) = \frac{\delta W[J, K]}{\delta J_\Phi(x)} \quad (3.340)$$

and Legendre transform only with respect to the ordinary sources J , not with respect to the BRST sources K :

$$\Gamma[\Phi, K] = W[J, K] - \int d^4x J_\Phi \Phi. \quad (3.341)$$

This gives

$$\frac{\delta \Gamma}{\delta \Phi} = -J_\Phi. \quad (3.342)$$

It also gives

$$\frac{\delta\Gamma}{\delta K_\Phi} = \frac{\delta W}{\delta K_\Phi}. \quad (3.343)$$

Let us check the second relation explicitly. When differentiating Γ with respect to K_Φ at fixed classical fields, the source J may depend on K :

$$\frac{\delta\Gamma}{\delta K_\Phi} = \frac{\delta W}{\delta K_\Phi} + \int d^4x \frac{\delta W}{\delta J_\Psi} \frac{\delta J_\Psi}{\delta K_\Phi} - \int d^4x \frac{\delta J_\Psi}{\delta K_\Phi} \Psi = \frac{\delta W}{\delta K_\Phi}, \quad (3.344)$$

because $\delta W/\delta J_\Psi = \Psi$. Therefore the two extra terms cancel.

Next perform the BRST change of variables inside the path integral,

$$\Phi \longrightarrow \Phi + \epsilon Q\Phi. \quad (3.345)$$

If the measure and the regularization preserve BRST symmetry, the integral does not change. The extended action is invariant because

$$QS_{\text{FP}} = 0, \quad Q(K_\Phi Q\Phi) = K_\Phi Q^2\Phi = 0, \quad (3.346)$$

where the K 's are fixed external sources and $Q^2 = 0$. Therefore the only variation comes from the ordinary source term that changes as

$$\begin{aligned} \int d^4x J_\Phi \Phi &\longrightarrow \int d^4x J_\Phi (\Phi + \epsilon Q\Phi) \\ &= \int d^4x J_\Phi \Phi + \epsilon \int d^4x J_\Phi Q\Phi. \end{aligned} \quad (3.347)$$

Thus the exponential in $Z[J, K]$ changes, to first order in ϵ , by

$$\delta \exp \left\{ iS_{\text{ext}} + i \int d^4x J_\Phi \Phi \right\} = i\epsilon \left(\int d^4x J_\Phi Q\Phi \right) \exp \left\{ iS_{\text{ext}} + i \int d^4x J_\Phi \Phi \right\}. \quad (3.348)$$

But the whole path integral cannot change under a change of integration variables, so $\delta Z = 0$. Therefore

$$0 = \delta Z = i\epsilon \int \mathcal{D}\Phi \left(\int d^4x J_\Phi Q\Phi \right) \exp \left\{ iS_{\text{ext}} + i \int d^4x J_\Phi \Phi \right\}. \quad (3.349)$$

Now divide by $Z[J, K]$ and remove the common factor $i\epsilon$. We get

$$0 = \int d^4x J_\Phi \langle Q\Phi \rangle_{J,K}. \quad (3.350)$$

Expanding the shorthand over all fields gives

$$0 = \int d^4x \left[J_A^{a\mu} \langle QA_\mu^a \rangle_{J,K} + J_c^a \langle Qc^a \rangle_{J,K} + J_\psi \langle Q\psi \rangle_{J,K} + J_{\bar{\psi}} \langle Q\bar{\psi} \rangle_{J,K} + J_c^a \langle Q\bar{c}^a \rangle_{J,K} + J_B^a \langle QB^a \rangle_{J,K} \right]. \quad (3.351)$$

Using $Q\bar{c}^a = B^a$ and $QB^a = 0$, this becomes

$$0 = \int d^4x \left[J_A^{a\mu} \langle QA_\mu^a \rangle_{J,K} + J_c^a \langle Qc^a \rangle_{J,K} + J_\psi \langle Q\psi \rangle_{J,K} + J_{\bar{\psi}} \langle Q\bar{\psi} \rangle_{J,K} + J_c^a \langle B^a \rangle_{J,K} \right]. \quad (3.352)$$

Finally, the reason for introducing the K -sources was precisely that

$$\frac{\delta W}{\delta K_A^{a\mu}} = \langle Q A_\mu^a \rangle_{J,K}, \quad \frac{\delta W}{\delta K_c^a} = \langle Q c^a \rangle_{J,K}, \quad \frac{\delta W}{\delta K_\psi} = \langle Q \psi \rangle_{J,K}, \quad \frac{\delta W}{\delta K_{\bar{\psi}}} = \langle Q \bar{\psi} \rangle_{J,K}. \quad (3.353)$$

Substituting these identities into the previous equation gives

$$0 = \int d^4x \left[J_A^{a\mu} \frac{\delta W}{\delta K_A^{a\mu}} + J_c^a \frac{\delta W}{\delta K_c^a} + J_\psi \frac{\delta W}{\delta K_\psi} + J_{\bar{\psi}} \frac{\delta W}{\delta K_{\bar{\psi}}} + J_{\bar{c}}^a \langle B^a \rangle_{J,K} \right], \quad (3.354)$$

up to the standard Grassmann signs. This is the BRST Ward identity for W .

Finally substitute (3.342) and (3.343) into (3.354). After multiplying the whole equation by an irrelevant minus sign, we obtain the Slavnov–Taylor identity for the effective action:

$$\mathcal{S}(\Gamma) = \int d^4x \left[\frac{\delta \Gamma}{\delta K_A^{a\mu}} \frac{\delta \Gamma}{\delta A_\mu^a} + \frac{\delta \Gamma}{\delta K_c^a} \frac{\delta \Gamma}{\delta c^a} + \frac{\delta \Gamma}{\delta K_\psi} \frac{\delta \Gamma}{\delta \psi} + \frac{\delta \Gamma}{\delta K_{\bar{\psi}}} \frac{\delta \Gamma}{\delta \bar{\psi}} + B^a \frac{\delta \Gamma}{\delta \bar{c}^a} \right] = 0. \quad (3.355)$$

This derivation shows the important point: S_{ext} does not “turn into” Γ . Rather, S_{ext} defines the path integral, the BRST change of variables gives the identity for $W[J, K]$, and the Legendre transform rewrites that identity as an identity for $\Gamma[\Phi, K]$.

At tree level the effective action is the extended classical action,

$$\Gamma^{(0)} = S_{\text{ext}}. \quad (3.356)$$

Therefore, when we study the divergent part at one fixed loop order, we may write schematically

$$\Gamma = S_{\text{ext}} + \Delta \quad (3.357)$$

where Δ stands for the local divergent functional defined above. We have suppressed finite terms and higher-loop terms, because they are not needed for the first-order constraint on the divergent part. Now expand the Slavnov–Taylor identity to first order in Δ :

$$\mathcal{S}(S_{\text{ext}} + \Delta) = \mathcal{S}(S_{\text{ext}}) + \mathcal{S}_{S_{\text{ext}}} \Delta + \mathcal{O}(\Delta^2). \quad (3.358)$$

This first-order operator $\mathcal{S}_{S_{\text{ext}}}$ is the linearized Slavnov–Taylor operator. In the notation of the discussion above we often write it as $\mathcal{S}_{\text{FP}}$, because after setting the external sources K to zero it is the BRST operator associated with the Faddeev–Popov action.

To see the linearization explicitly, look only at the gauge-field part:

$$\begin{aligned} \frac{\delta \Gamma}{\delta K_A^{a\mu}} \frac{\delta \Gamma}{\delta A_\mu^a} &= \left(\frac{\delta S_{\text{ext}}}{\delta K_A^{a\mu}} + \frac{\delta \Delta}{\delta K_A^{a\mu}} \right) \left(\frac{\delta S_{\text{ext}}}{\delta A_\mu^a} + \frac{\delta \Delta}{\delta A_\mu^a} \right) \\ &= \underbrace{\frac{\delta S_{\text{ext}}}{\delta K_A^{a\mu}} \frac{\delta S_{\text{ext}}}{\delta A_\mu^a}}_{\mathcal{S}(S_{\text{ext}})} + \underbrace{\frac{\delta S_{\text{ext}}}{\delta K_A^{a\mu}} \frac{\delta \Delta}{\delta A_\mu^a} + \frac{\delta \Delta}{\delta K_A^{a\mu}} \frac{\delta S_{\text{ext}}}{\delta A_\mu^a}}_{\mathcal{S}_{S_{\text{ext}}(A)} \Delta} + \mathcal{O}(\Delta^2). \end{aligned} \quad (3.359)$$

The first term belongs to $\mathcal{S}(S_{\text{ext}})$, which vanishes by classical BRST invariance. The remaining two terms are the linearized contribution. Doing the same expansion for every field gives

$$\begin{aligned} \mathcal{S}_{S_{\text{ext}}} \Delta &= \int d^4x \left[\frac{\delta S_{\text{ext}}}{\delta K_A^{a\mu}} \frac{\delta \Delta}{\delta A_\mu^a} + \frac{\delta \Delta}{\delta K_A^{a\mu}} \frac{\delta S_{\text{ext}}}{\delta A_\mu^a} + \frac{\delta S_{\text{ext}}}{\delta K_c^a} \frac{\delta \Delta}{\delta c^a} + \frac{\delta \Delta}{\delta K_c^a} \frac{\delta S_{\text{ext}}}{\delta c^a} \right. \\ &\quad \left. + \frac{\delta S_{\text{ext}}}{\delta K_\psi} \frac{\delta \Delta}{\delta \psi} + \frac{\delta \Delta}{\delta K_\psi} \frac{\delta S_{\text{ext}}}{\delta \psi} + \frac{\delta S_{\text{ext}}}{\delta K_{\bar{\psi}}} \frac{\delta \Delta}{\delta \bar{\psi}} + \frac{\delta \Delta}{\delta K_{\bar{\psi}}} \frac{\delta S_{\text{ext}}}{\delta \bar{\psi}} + B^a \frac{\delta \Delta}{\delta \bar{c}^a} \right], \end{aligned} \quad (3.360)$$

again suppressing only the Grassmann sign factors. This is the explicit linearized Slavnov–Taylor operator.

For the ordinary counterterms in the Lagrangian, such as $A_\mu^a A^{a\mu}$, $F_{\mu\nu}^a F^{a\mu\nu}$, or $\bar{\psi} i \not{D} \psi$, the candidate Δ does not contain the external sources K . Hence

$$\frac{\delta\Delta}{\delta K_A} = \frac{\delta\Delta}{\delta K_c} = \frac{\delta\Delta}{\delta K_\psi} = \frac{\delta\Delta}{\delta K_{\bar{\psi}}} = 0. \quad (3.361)$$

Using the definition of the K -sources, (3.360) then reduces to

$$\begin{aligned} \mathcal{S}_{S_{\text{ext}}} \Delta &= \int d^4x \left[Q A_\mu^a \frac{\delta\Delta}{\delta A_\mu^a} + Q c^a \frac{\delta\Delta}{\delta c^a} + Q \psi \frac{\delta\Delta}{\delta \psi} + Q \bar{\psi} \frac{\delta\Delta}{\delta \bar{\psi}} + B^a \frac{\delta\Delta}{\delta \bar{c}^a} \right] \\ &= \int d^4x \left[Q A_\mu^a \frac{\delta\Delta}{\delta A_\mu^a} + Q c^a \frac{\delta\Delta}{\delta c^a} + Q \bar{c}^a \frac{\delta\Delta}{\delta \bar{c}^a} + Q \psi \frac{\delta\Delta}{\delta \psi} + Q \bar{\psi} \frac{\delta\Delta}{\delta \bar{\psi}} \right], \end{aligned} \quad (3.362)$$

because $Q\bar{c}^a = B^a$. But the last line is exactly the BRST variation of the functional Δ :

$$Q\Delta = \int d^4x \sum_{\Phi=A,c,\bar{c},\psi,\bar{\psi}} Q\Phi \frac{\delta\Delta}{\delta\Phi}. \quad (3.363)$$

Therefore, for ordinary local counterterms which do not contain the external BRST sources,

$$\boxed{\mathcal{S}_{S_{\text{ext}}} \Delta = Q\Delta} \quad (3.364)$$

up to total derivatives and the standard Grassmann signs. This is why, in practice, we can test whether a candidate counterterm is allowed by simply applying the BRST transformations to it. If $Q\Delta = 0$, or if $Q\Delta$ is only a spacetime total derivative, then the term is compatible with the linearized Slavnov–Taylor identity. If not, it is forbidden.

Now (3.326) tells us (3.358) must satisfy

$$\mathcal{S}(S_{\text{ext}}) + \mathcal{S}_{S_{\text{ext}}} \Delta = 0. \quad (3.365)$$

Since the classical action is BRST invariant, we have

$$\mathcal{S}(S_{\text{ext}}) = 0. \quad (3.366)$$

Therefore, after lower-loop subdivergences have been subtracted, with the relation in (3.364) the remaining divergent part must satisfy

$$\mathcal{S}_{S_{\text{ext}}} \Gamma_{\text{div}}^{(n)} = Q\Gamma_{\text{div}}^{(n)} = 0. \quad (3.367)$$

This is the precise meaning of the sentence “the divergent part satisfies the linearized Slavnov–Taylor identity”. Less formally, it means that the divergent counterterm cannot be an arbitrary local expression; it must be compatible with BRST symmetry.

Combining this with power counting gives the following list of restrictions:

UV divergences must be local, have dimension no larger than four, have ghost number zero, and be Lorentz invariant, color invariant, and BRST invariant.

Let us justify each phrase one by one.

- **Local.** UV divergences come from the region where loop momentum is very large. In that region the graph can only depend on external momenta through a polynomial divergent part. A polynomial in external momenta is a local operator in position space.
- **Dimension no larger than four.** From the superficial degree of divergence computed above, only operators of mass dimension ≤ 4 can be needed as counterterms in four spacetime dimensions. Operators such as A^6 , $F_{\mu\nu}D^2F^{\mu\nu}$, or $(\bar{\psi}\psi)^2$ have dimension larger than four, so power counting already excludes them.
- **Ghost number zero.** The action and the effective action have ghost number zero. Hence a divergent counterterm must also have ghost number zero. For example, $\bar{c}c$ has ghost number $-1 + 1 = 0$, but a single c cannot appear by itself.
- **Lorentz invariance.** The original theory and dimensional regularization preserve Lorentz symmetry. Therefore all Lorentz indices must be properly contracted. For example, $F_{\mu\nu}^a F^{a\mu\nu}$ is allowed by Lorentz symmetry, while an expression with a free Lorentz index is not.
- **Color invariance.** The original action has global color invariance. Therefore color indices must also be contracted, for example by δ^{ab} or f^{abc} in the appropriate places.
- **BRST invariance.** This is the extra condition coming from the Slavnov–Taylor identity. It removes terms which are allowed by naive power counting but are incompatible with gauge symmetry.

For instance, a gauge-boson mass term has dimension two,

$$\frac{1}{2}M_A^2 A_\mu^a A^{a\mu}, \quad (3.368)$$

and it also has ghost number zero, Lorentz invariance, and color invariance. Thus, if we only used power counting, this term would look allowed. However, it is not BRST invariant. Indeed,

$$Q(A_\mu^a A^{a\mu}) = (QA_\mu^a)A^{a\mu} + A_\mu^a Q(A^{a\mu}) = 2A^{a\mu}D_\mu^{ab}c^b. \quad (3.369)$$

Expanding the covariant derivative makes the problem visible:

$$2A^{a\mu}D_\mu^{ab}c^b = 2A^{a\mu}\partial_\mu c^a + 2gf^{acb}A^{a\mu}A_\mu^c c^b. \quad (3.370)$$

This is not zero, and it is not merely a total derivative. Therefore a mass term for the gauge boson violates the linearized Slavnov–Taylor identity. In other words, BRST symmetry forbids it as a counterterm.

The remaining allowed divergent local terms are precisely those that can be absorbed into the normalization of the fields and parameters already present in $\mathcal{L}_{\text{FP}}$:

$$A_\mu, \quad \psi, \quad c, \quad g, \quad m \quad (3.371)$$

plus the gauge-fixing parameter ξ if one does not stay in Feynman gauge.

Equivalently, the most general allowed local divergent structure has the same form as

$$-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}, \quad \bar{\psi}i\not{D}\psi, \quad m\bar{\psi}\psi, \quad \bar{c}^a(-\partial^\mu D_\mu^{ab})c^b, \quad -\frac{1}{2\xi}(\partial^\mu A_\mu^a)^2. \quad (3.372)$$

Therefore no new independent interaction has to be introduced. This is the meaning of perturbative renormalizability of the Faddeev–Popov theory.

4.2 Counter terms of Faddeev–Popov Lagrangian

Now we have prepared to write the renormalization relates bare quantities to renormalized ones for the three type fields ψ , A_μ^a and c^a :

$$\psi_0 = Z_2^{1/2}\psi, \quad \bar{\psi}_0 = Z_2^{1/2}\bar{\psi}, \quad (3.373)$$

$$c_0^a = (Z_2^c)^{1/2}c^a, \quad \bar{c}_0^a = (Z_2^c)^{1/2}\bar{c}^a, \quad (3.374)$$

$$A_{0,\mu}^a = Z_3^{1/2}A_\mu^a, \quad (3.375)$$

where fields renormalization can be expanded to get counter terms:

$$Z_2 = 1 + \delta_2, \quad Z_3 = 1 + \delta_3, \quad Z_2^c = 1 + \delta_2^c. \quad (3.376)$$

The mass and the several vertex renormalizations can be seen by putting the field renormalization factors into the Lagrangian.

Fermion mass $m\bar{\psi}\psi$

First the bare mass term becomes

$$m_0\bar{\psi}_0\psi_0 = m_0(Z_2^{1/2}\bar{\psi})(Z_2^{1/2}\psi) = Z_2m_0\bar{\psi}\psi. \quad (3.377)$$

Thus the coefficient multiplying $\bar{\psi}\psi$ is defined as

$$Z_2m_0 = m + \delta m, \quad (3.378)$$

so the original mass term becomes the renormalized mass term plus the mass counterterm,

$$\boxed{Z_2m_0\bar{\psi}\psi = m\bar{\psi}\psi + \delta m\bar{\psi}\psi.} \quad (3.379)$$

Similarly, each interaction vertex gives one coefficient relation.

Fermion–gauge-boson vertex $A\bar{\psi}\psi$.

This vertex comes from the fermion covariant-derivative term $\bar{\psi}_0 i \not{D}_0 \psi_0$. Its interaction part is

$$\begin{aligned} \mathcal{L}_{A\bar{\psi}\psi}^{(0)} &= g_0 A_{0,\mu}^a \bar{\psi}_0 \gamma^\mu t^a \psi_0 \\ &= g_0 Z_2 Z_3^{1/2} A_\mu^a \bar{\psi} \gamma^\mu t^a \psi. \end{aligned} \quad (3.380)$$

We define the renormalized coefficient of the same operator by

$$\mathcal{L}_{A\bar{\psi}\psi} = g Z_1 A_\mu^a \bar{\psi} \gamma^\mu t^a \psi. \quad (3.381)$$

Therefore

$$\boxed{Z_2 Z_3^{1/2} g_0 = g Z_1 = g(1 + \delta_1).} \quad (3.382)$$

Three-gauge-boson vertex AAA .

This vertex comes from the Yang–Mills kinetic term $-\frac{1}{4}F_{0,\mu\nu}^a F_0^{a\mu\nu}$. The part with three gauge fields is

$$\begin{aligned}\mathcal{L}_{AAA}^{(0)} &= -g_0 f^{abc} (\partial_\mu A_{0,\nu}^a) A_0^{b\mu} A_0^{c\nu} \\ &= -g_0 Z_3^{3/2} f^{abc} (\partial_\mu A_\nu^a) A^{b\mu} A^{c\nu}.\end{aligned}\quad (3.383)$$

We write the renormalized coefficient as

$$\mathcal{L}_{AAA} = -g Z_1^{3g} f^{abc} (\partial_\mu A_\nu^a) A^{b\mu} A^{c\nu}. \quad (3.384)$$

Therefore

$$\boxed{Z_3^{3/2} g_0 = g Z_1^{3g} = g(1 + \delta_1^{3g}).} \quad (3.385)$$

Four-gauge-boson vertex $AAAA$.

This vertex also comes from $-\frac{1}{4}F_{0,\mu\nu}^a F_0^{a\mu\nu}$. The part with four gauge fields is

$$\begin{aligned}\mathcal{L}_{AAAA}^{(0)} &= -\frac{1}{4}g_0^2 (f^{abc} A_{0,\mu}^b A_{0,\nu}^c) (f^{ade} A_0^{d\mu} A_0^{e\nu}) \\ &= -\frac{1}{4}g_0^2 Z_3^2 (f^{abc} A_\mu^b A_\nu^c) (f^{ade} A^{d\mu} A^{e\nu}).\end{aligned}\quad (3.386)$$

We write the renormalized coefficient as

$$\mathcal{L}_{AAAA} = -\frac{1}{4}g^2 Z_1^{4g} (f^{abc} A_\mu^b A_\nu^c) (f^{ade} A^{d\mu} A^{e\nu}). \quad (3.387)$$

Therefore

$$\boxed{Z_3^2 g_0^2 = g^2 Z_1^{4g} = g^2(1 + \delta_1^{4g}).} \quad (3.388)$$

Ghost–gauge-boson vertex $A\bar{c}c$.

This vertex comes from the ghost term $\bar{c}_0^a (-\partial^\mu D_{0,\mu}^{ab}) c_0^b$. After integrating by parts, its interaction part can be written as

$$\begin{aligned}\mathcal{L}_{A\bar{c}c}^{(0)} &= g_0 f^{abc} (\partial^\mu \bar{c}_0^a) A_{0,\mu}^b c_0^c \\ &= g_0 Z_2 Z_3^{1/2} f^{abc} (\partial^\mu \bar{c}^a) A_\mu^b c^c.\end{aligned}\quad (3.389)$$

We write the renormalized coefficient as

$$\mathcal{L}_{A\bar{c}c} = g Z_1^c f^{abc} (\partial^\mu \bar{c}^a) A_\mu^b c^c. \quad (3.390)$$

Therefore

$$\boxed{Z_2^c Z_3^{1/2} g_0 = g Z_1^c = g(1 + \delta_1^c).} \quad (3.391)$$

Thus the counterterm Lagrangian can be written as

$$\begin{aligned}\mathcal{L}_{\text{ct}} &= \bar{\psi} (i\delta_2 \not{\partial} - \delta m) \psi + g\delta_1 A_\mu^a \bar{\psi} \gamma^\mu t^a \psi \\ &\quad - \frac{1}{4}\delta_3 (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a)^2 - g\delta_1^{3g} f^{abc} (\partial_\mu A_\nu^a) A^{b\mu} A^{c\nu} - \frac{1}{4}g^2 \delta_1^{4g} (f^{abc} A_\mu^b A_\nu^c) (f^{ade} A^{d\mu} A^{e\nu}) \\ &\quad - \delta_2^c \bar{c}^a \partial_\mu \partial^\mu c^a + g\delta_1^c f^{abc} (\partial^\mu \bar{c}^a) A_\mu^b c^c.\end{aligned}\quad (3.392)$$

Proof

We begin with the Slavnov-Taylor identity:

$$\mathcal{S}(\Gamma) = \int d^4x \left[\frac{\delta\Gamma}{\delta K_A^{a\mu}} \frac{\delta\Gamma}{\delta A_\mu^a} + \frac{\delta\Gamma}{\delta K_c^a} \frac{\delta\Gamma}{\delta c^a} + B^a \frac{\delta\Gamma}{\delta \bar{c}^a} + \dots \right] = 0. \quad (3.402)$$

Now take derivative to it:

$$\frac{\delta^2}{\delta c^b(y) \delta A_\nu^c(z)} \mathcal{S}(\Gamma) = 0. \quad (3.403)$$

And setting $A = c = \bar{c} = \psi = \bar{\psi} = B = K = 0$. So most of terms will vanish because of $\frac{\delta\Gamma}{\delta K_\Phi} = \langle Q\Phi \rangle_{J,K}$, only left

$$\int d^4x \frac{\delta^2\Gamma}{\delta K_A^{a\mu} \delta c^b} \frac{\delta^2\Gamma}{\delta A_\mu^a \delta A_\nu^c} = 0. \quad (3.404)$$

Notice that for $A = 0$, we have

$$QA_\mu^a = (D_\mu c)^a = \partial_\mu c^a + \underbrace{gf^{abc} A_\mu^b c^c}_0. \quad (3.405)$$

Thus at tree level, we have

$$\Gamma^{(0)} = S_{\text{ext}} \supset \int d^4x K_A^{a\mu}(x) [\partial_\mu c^a(x) + gf^{ade} A_\mu^d(x) c^e(x)], \quad (3.406)$$

thus

$$\Gamma_{K_A^{a\mu} c^b}^{(0)}(x, y) \equiv \frac{\delta^2\Gamma}{\delta K_A^{a\mu} \delta c^b} = \delta^{ab} \partial_\mu^x \delta(x - y). \quad (3.407)$$

In the momentum space, we have

$$\Gamma_{K_A^{a\mu} c^b}^{(0)}(q) = i\delta^{ab} q_\mu. \quad (3.408)$$

The loop correction is merely contribute a scalar function $G(p^2) = 1 + \mathcal{O}(\hbar)$, thus it becomes

$$\Gamma_{K_A^{a\mu} c^b}(q) = iq_\mu \delta^{ab} G(q^2). \quad (3.409)$$

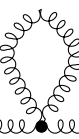
Put it back to (3.404), we then have

$$iq^\mu G(q^2) \Gamma_{A_\mu^b A_\nu^c}(q) = 0. \quad (3.410)$$

Because $G(p^2) \neq 0$, $\Gamma_{A_\mu^a A_\nu^b} = \Gamma_{A_\mu^a A_\nu^b}^{(0)} + \Pi_{\mu\nu}^{ab} + \dots$ and the tree level part originally satisfies $q^\mu \Gamma_{A_\mu^a A_\nu^b}^{(0)} = 0$, thus

$$q^\mu \Pi_{\mu\nu}^{ab} = 0. \quad (3.411)$$

Base on th Feynman rules, there should be four 1-loop correction diagrams plus a counter term graph. However the massless gluon tadpole



$$a, \mu \text{ --- } \bullet \text{ --- } b, \nu \quad (3.412)$$

vanishes in dimensional regularization since the vertex Feynman rule doesn't contain any momentum term, so the integral is:

$$\int \frac{d^d k}{(2\pi)^d} \frac{g_{\mu\nu}}{k^2} \delta^{ab}. \quad (3.413)$$

This kind of integral is called **scaleless**, because it doesn't involve any external momentum or mass, so under the rescaling $k \rightarrow \lambda k$, it has

$$d^d k \rightarrow \lambda^d d^d k, \quad \frac{1}{k^2} \rightarrow \lambda^{-2} \frac{1}{k^2}. \quad (3.414)$$

This leads to

$$I \rightarrow \lambda^{d-2} I, \quad (3.415)$$

which means the integral relies on the arbitrary rescaling parameter. This is unphysical unless $I = 0$. Or we can also explain it by direct calculation. Ignore the harmless tensor and color factors $g_{\mu\nu}\delta^{ab}$ and define the Euclidean scalar integral

$$I = \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2}. \quad (3.416)$$

The factor μ^{4-d} only keeps the dimension of the integral fixed when $d \neq 4$. After angular integration,

$$I = \mu^{4-d} \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty dk k^{d-3}, \quad \Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \quad (3.417)$$

Now introduce an arbitrary separating scale Λ and split the radial integral into an infrared part and an ultraviolet part:

$$\int_0^\infty dk k^{d-3} = \int_0^\Lambda dk k^{d-3} + \int_\Lambda^\infty dk k^{d-3}. \quad (3.418)$$

The first integral is convergent for $\text{Re } d > 2$ and gives

$$I_{\text{IR}} = \int_0^\Lambda dk k^{d-3} = \frac{\Lambda^{d-2}}{d-2}. \quad (3.419)$$

The second integral is convergent for $\text{Re } d < 2$ and gives

$$I_{\text{UV}} = \int_\Lambda^\infty dk k^{d-3} = -\frac{\Lambda^{d-2}}{d-2}. \quad (3.420)$$

These two convergence regions do not overlap. Dimensional regularization means that we regard these answers as analytic functions of d and then continue them to the same value of d . Adding the two analytically continued pieces,

$$I = \mu^{4-d} \frac{\Omega_{d-1}}{(2\pi)^d} \left(\frac{\Lambda^{d-2}}{d-2} - \frac{\Lambda^{d-2}}{d-2} \right) = 0. \quad (3.421)$$

Thus the massless tadpole is zero in dimensional regularization:

$$\boxed{\int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2} = 0.} \quad (3.422)$$

Generally speaking, we have the conclusion: **scaleless integral is zero in dimensional regularization.**

Fermion loop.

For one fermion flavor, the closed fermion loop gives the color trace

$$(t^a)_{ij}(t^b)_{ji} = \text{tr}(t^a t^b), \quad (3.432)$$

and the closed spinor indices give the Dirac trace

$$(\gamma_\mu)_{\alpha\beta}(\not{K})_{\beta\gamma}(\gamma_\nu)_{\gamma\delta}(\not{K} + \not{q})_{\delta\alpha} = \text{tr}[\gamma_\mu \not{K} \gamma_\nu (\not{K} + \not{q})]. \quad (3.433)$$

With n_f flavors,

$$\begin{aligned} \Pi_{(f)\mu\nu}^{ab}(q) &= -n_f g^2 \text{tr}(t^a t^b) \int \frac{d^d k}{(2\pi)^d} \frac{\text{tr}[\gamma_\mu \not{K} \gamma_\nu (\not{K} + \not{q})]}{k^2 (k + q)^2} \\ &= -n_f g^2 C(r) \delta^{ab} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\mu\nu}^{(f)}}{k^2 (k + q)^2}, \end{aligned} \quad (3.434)$$

where the color trace gives the fundamental representation of $SU(N)$ gauge group, namely: $\text{tr}(t^a t^b) = C(r) \delta^{ab}$, and $C(r) = T_F = \frac{1}{2}$. The Dirac trace is

$$\begin{aligned} N_{\mu\nu}^{(f)} &= \text{tr}[\gamma_\mu \not{K} \gamma_\nu (\not{K} + \not{q})] \\ &= 4[k_\mu (k + q)_\nu + k_\nu (k + q)_\mu - g_{\mu\nu} k \cdot (k + q)]. \end{aligned} \quad (3.435)$$

Now introduce the Feynman parameter and shift

$$k = \ell - xq, \quad k + q = \ell + (1 - x)q. \quad (3.436)$$

Then

$$\begin{aligned} k_\mu (k + q)_\nu &= \ell_\mu \ell_\nu + (1 - x)\ell_\mu q_\nu - xq_\mu \ell_\nu - x(1 - x)q_\mu q_\nu, \\ k_\nu (k + q)_\mu &= \ell_\nu \ell_\mu + (1 - x)\ell_\nu q_\mu - xq_\nu \ell_\mu - x(1 - x)q_\nu q_\mu, \\ k \cdot (k + q) &= \ell^2 + (1 - 2x)\ell \cdot q - x(1 - x)q^2. \end{aligned} \quad (3.437)$$

All terms odd in ℓ vanish after integration. Therefore the numerator can be replaced, under the integral, by

$$N_{\mu\nu}^{(f)} \longrightarrow 4[2\ell_\mu \ell_\nu - g_{\mu\nu} \ell^2 - 2x(1 - x)q_\mu q_\nu + x(1 - x)g_{\mu\nu} q^2]. \quad (3.438)$$

Now contract with the transverse projector. Since

$$P^{\mu\nu} q_\mu q_\nu = 0, \quad P^{\mu\nu} g_{\mu\nu} = d - 1, \quad P^{\mu\nu} \ell_\mu \ell_\nu \rightarrow \frac{d - 1}{d} \ell^2, \quad (3.439)$$

we get

$$P^{\mu\nu} N_{\mu\nu}^{(f)} \longrightarrow 4(d - 1) \left[\left(\frac{2}{d} - 1 \right) \ell^2 + x(1 - x)q^2 \right]. \quad (3.440)$$

Using the ℓ^2 integral above with $\Delta = -x(1 - x)q^2$, the divergent part inside the square bracket becomes

$$\left(\frac{2}{d} - 1 \right) \left[-\frac{d}{d - 2} \Delta \right] + x(1 - x)q^2 = 2x(1 - x)q^2. \quad (3.441)$$

Thus

$$P^{\mu\nu} N_{\mu\nu}^{(f)} \longrightarrow 8(d-1)x(1-x)q^2. \quad (3.442)$$

After dividing by the projector normalization $i(d-1)q^2\delta^{ab}$, the divergent part becomes

$$\Pi_{(f)}(q^2) = -\frac{g^2}{(4\pi)^2} n_f C(r) \Gamma(\epsilon) (-q^2)^{-\epsilon} \int_0^1 dx 8x(1-x) + \text{finite terms}. \quad (3.443)$$

The remaining x integral is elementary:

$$\int_0^1 dx 8x(1-x) = 8 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{4}{3}. \quad (3.444)$$

Therefore

$$\Pi_{(f)}(q^2) = -\frac{g^2}{(4\pi)^2} \frac{4}{3} n_f C(r) \Gamma(\epsilon) (-q^2)^{-\epsilon} + \text{finite terms}. \quad (3.445)$$

Ghost loop.

The ghost loop also carries a minus sign because ghosts are Grassmann fields. Using the adjoint representation relation $f^{acd}f^{bcd} = C_2(G)\delta^{ab}$, the diagram is

$$\Pi_{(c)\mu\nu}^{ab}(q) = -g^2 C_2(G) \delta^{ab} \int \frac{d^d k}{(2\pi)^d} \frac{k_\mu(k+q)_\nu}{k^2(k+q)^2}. \quad (3.446)$$

Again set

$$k = \ell - xq, \quad k + q = \ell + (1-x)q. \quad (3.447)$$

Then the numerator is

$$k_\mu(k+q)_\nu = \ell_\mu \ell_\nu + (1-x)\ell_\mu q_\nu - xq_\mu \ell_\nu - x(1-x)q_\mu q_\nu. \quad (3.448)$$

The terms linear in ℓ integrate to zero, and the $q_\mu q_\nu$ term is killed by the transverse projector. Hence only the $\ell_\mu \ell_\nu$ term contributes:

$$P^{\mu\nu} k_\mu(k+q)_\nu \longrightarrow P^{\mu\nu} \ell_\mu \ell_\nu \rightarrow \frac{d-1}{d} \ell^2. \quad (3.449)$$

Using the logarithmically divergent part of the ℓ^2 integral and then dividing by $i(d-1)q^2\delta^{ab}$ gives the scalar coefficient

$$\frac{1}{(d-1)q^2} \frac{d-1}{d} \left[-\frac{d}{d-2} \Delta \right] = \frac{x(1-x)}{d-2} = \frac{1}{2} x(1-x) + \mathcal{O}(\epsilon). \quad (3.450)$$

With the standard ghost-gluon vertex convention used in the Feynman rule, the two momentum factors give an extra factor 4. Thus the divergent contribution is equivalently written as

$$\Pi_{(c)}(q^2) = -\frac{g^2}{(4\pi)^2} C_2(G) \Gamma(\epsilon) (-q^2)^{-\epsilon} \int_0^1 dx 2x(1-x) + \text{finite terms}. \quad (3.451)$$

Since

$$\int_0^1 dx 2x(1-x) = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}, \quad (3.452)$$

the ghost loop contributes

$$\Pi_{(c)}(q^2) = -\frac{g^2}{(4\pi)^2} \frac{1}{3} C_2(G) \Gamma(\epsilon) (-q^2)^{-\epsilon} + \text{finite terms}. \quad (3.453)$$

Gluon loop.

The gluon loop uses two three-gauge-boson vertices and has a symmetry factor 1/2. With all momenta incoming, write the three-gauge-boson vertex without the overall factor gf^{abc} as

$$V_{\alpha\beta\gamma}(p, q, r) = g_{\alpha\beta}(p - q)_\gamma + g_{\beta\gamma}(q - r)_\alpha + g_{\gamma\alpha}(r - p)_\beta. \quad (3.454)$$

The diagram is therefore

$$\begin{aligned} \Pi_{(g)\mu\nu}^{ab}(q) &= \frac{1}{2}g^2 f^{acd} f^{bcd} \int \frac{d^d k}{(2\pi)^d} \frac{V_{\mu\rho\sigma}(q, k, -k - q) V_\nu^{\rho\sigma}(-q, k + q, -k)}{k^2(k + q)^2} \\ &= \frac{1}{2}g^2 C_2(G) \delta^{ab} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\mu\nu}^{(g)}}{k^2(k + q)^2}. \end{aligned} \quad (3.455)$$

For the numerator, first expand the two vertices. One finds

$$\begin{aligned} V_{\mu\rho\sigma}(q, k, -k - q) &= g_{\mu\rho}(q - k)_\sigma + g_{\rho\sigma}(2k + q)_\mu - g_{\sigma\mu}(k + 2q)_\rho, \\ V_\nu^{\rho\sigma}(-q, k + q, -k) &= \delta_\nu^\rho(-k - 2q)^\sigma + g^{\rho\sigma}(2k + q)_\nu + \delta_\nu^\sigma(q - k)^\rho. \end{aligned} \quad (3.456)$$

Therefore

$$N_{\mu\nu}^{(g)} = V_{\mu\rho\sigma}(q, k, -k - q) V_\nu^{\rho\sigma}(-q, k + q, -k). \quad (3.457)$$

Multiplying the two brackets and using $g_{\rho\sigma}g^{\rho\sigma} = d$ gives

$$\begin{aligned} N_{\mu\nu}^{(g)} &= (4d - 6)k_\mu k_\nu + (2d - 3)(k_\mu q_\nu + q_\mu k_\nu) \\ &\quad + (d + 3)q_\mu q_\nu + 2g_{\mu\nu}(k^2 + k \cdot q - 2q^2). \end{aligned} \quad (3.458)$$

Now set $k = \ell - xq$. Odd powers of ℓ vanish. After the shift, the even part can be written as

$$\begin{aligned} N_{\mu\nu}^{(g)} &\longrightarrow (4d - 6)\ell_\mu \ell_\nu + 2g_{\mu\nu}\ell^2 \\ &\quad + x(1 - x)[-2(4d - 6)q_\mu q_\nu + 4g_{\mu\nu}q^2] + \text{terms killed by } P^{\mu\nu}. \end{aligned} \quad (3.459)$$

Applying the transverse projector and reducing $\ell_\mu \ell_\nu \rightarrow g_{\mu\nu}\ell^2/d$ gives

$$P^{\mu\nu} N_{\mu\nu}^{(g)} \longrightarrow (d - 1) \left[\left(\frac{4d - 6}{d} + 2 \right) \ell^2 + 4x(1 - x)q^2 \right]. \quad (3.460)$$

Using the same ℓ^2 integral relation with $\Delta = -x(1 - x)q^2$, the divergent part of the bracket becomes

$$24x(1 - x)q^2. \quad (3.461)$$

The diagram has the symmetry factor 1/2, so after dividing by $i(d - 1)q^2\delta^{ab}$ the scalar integrand is

$$\frac{1}{2} 24x(1 - x) = 12x(1 - x). \quad (3.462)$$

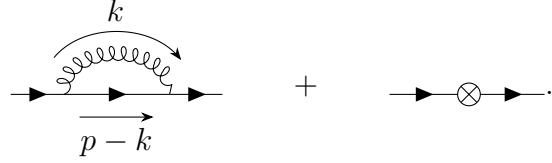
Thus the divergent scalar part is

$$\Pi_{(g)}(q^2) = \frac{g^2}{(4\pi)^2} C_2(G) \Gamma(\epsilon) (-q^2)^{-\epsilon} \int_0^1 dx 12x(1 - x) + \text{finite terms}. \quad (3.463)$$

The parameter integral is

$$\int_0^1 dx 12x(1 - x) = 12 \left(\frac{1}{2} - \frac{1}{3} \right) = 2. \quad (3.464)$$

At one loop the non-vanishing diagrams are the gauge-boson loop and the fermion-field counterterm:



$$(3.475)$$

We keep only the coefficient of $\not{p}$, because that is the part cancelled by the wave-function counterterm δ_2 . In Feynman gauge, using $(t^a t^a)_{ij} = C_2(r)\delta_{ij}$, the loop contribution is

$$i\Sigma_{\text{loop}}(p) = ig^2 C_2(r) \int \frac{d^d k}{(2\pi)^d} \frac{\gamma^\mu (\not{p} - \not{k}) \gamma_\mu}{k^2 (p-k)^2}. \quad (3.476)$$

The numerator is reduced by

$$\gamma^\mu \gamma^\alpha \gamma_\mu = (2-d)\gamma^\alpha, \quad \gamma^\mu (\not{p} - \not{k}) \gamma_\mu = (2-d)(\not{p} - \not{k}). \quad (3.477)$$

Combine the two denominators:

$$\begin{aligned} \frac{1}{k^2 (p-k)^2} &= \int_0^1 dx \frac{1}{[xk^2 + (1-x)(p-k)^2]^2} \\ &= \int_0^1 dx \frac{1}{[\ell^2 + x(1-x)p^2]^2}, \quad \ell = k - (1-x)p. \end{aligned} \quad (3.478)$$

Since

$$p - k = xp - \ell, \quad (3.479)$$

the odd term proportional to ℓ integrates to zero:

$$i\Sigma_{\text{loop}}(p) = ig^2 C_2(r) (2-d) \not{p} \int_0^1 dx x \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[\ell^2 + x(1-x)p^2]^2} + \text{finite terms}. \quad (3.480)$$

Using the same logarithmically divergent scalar integral as above,

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[\ell^2 + x(1-x)p^2]^2} = \frac{i}{(4\pi)^2} \Gamma(\epsilon) (-p^2)^{-\epsilon} + \text{finite terms}, \quad (3.481)$$

and the x integral comes only from the even part xp in $p - k = xp - \ell$:

$$(2-d) \int_0^1 dx x = (2-d) \frac{1}{2} = -1 + \epsilon, \quad (3.482)$$

so the divergent coefficient is simply -1 at leading order in ϵ . the divergent part has the form

$$\Sigma_{\text{loop}}(p) = i \frac{g^2}{(4\pi)^2} C_2(r) \not{p} \Gamma(\epsilon) (-p^2)^{-\epsilon} + \text{finite terms}. \quad (3.483)$$

The counterterm $\bar{\psi} i \delta_2 \not{\partial} \psi$ contributes

$$\Sigma_{\text{ct}}(p) = \delta_2 \not{p}. \quad (3.484)$$

Therefore

$$\Sigma(p) = i \frac{g^2}{(4\pi)^2} C_2(r) \not{p} \Gamma(\epsilon) (-p^2)^{-\epsilon} + \delta_2 \not{p} + \text{finite terms}. \quad (3.485)$$

so the Dirac numerator becomes

$$\begin{aligned}
k_\alpha k_\beta \gamma^\rho \gamma^\alpha \gamma^\mu \gamma^\beta \gamma_\rho &\longrightarrow \frac{k^2}{d} \gamma^\rho \gamma^\alpha \gamma^\mu \gamma_\alpha \gamma_\rho \\
&= \frac{k^2}{d} (2-d) \gamma^\rho \gamma^\mu \gamma_\rho \\
&= \frac{k^2}{d} (2-d)^2 \gamma^\mu \\
&= k^2 \gamma^\mu + \mathcal{O}(\epsilon).
\end{aligned} \tag{3.494}$$

Meanwhile the three denominators behave in the logarithmic part as

$$k^2(p' - k)^2(p - k)^2 \longrightarrow (k^2)^3. \tag{3.495}$$

Therefore the UV divergent part reduces to the standard logarithmic integral

$$\int \frac{d^d k}{(2\pi)^d} \frac{k^2}{(k^2)^3} = \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2)^2} \longrightarrow \frac{i}{(4\pi)^2} \Gamma(\epsilon) (-q^2)^{-\epsilon}. \tag{3.496}$$

Combining this with $t^b t^a t^b = (C_2(r) - \frac{1}{2} C_2(G)) t^a$, the logarithmically divergent part is

$$\Lambda_{\text{A,div}}^\mu = i g \gamma^\mu (t^a)_{ij} \frac{g^2}{(4\pi)^2} \left(C_2(r) - \frac{1}{2} C_2(G) \right) \Gamma(\epsilon) (-q^2)^{-\epsilon}. \tag{3.497}$$

Non-Abelian diagram

The non-Abelian diagram has the characteristic color factor

$$\begin{aligned}
f^{abc}(t^b)_{i\ell}(t^c)_{\ell j} &= \frac{1}{2} f^{abc} ([t^b, t^c] + \{t^b, t^c\})_{ij} \\
&= \frac{1}{2} f^{abc} [t^b, t^c]_{ij} \\
&= \frac{i}{2} f^{abc} f^{bcd} (t^d)_{ij} = \frac{i}{2} C_2(G) (t^a)_{ij}.
\end{aligned} \tag{3.498}$$

Its loop integral is

$$\Lambda_{\text{NA}}^\mu = i g^3 f^{abc} (t^b t^c)_{ij} \int \frac{d^d k}{(2\pi)^d} \frac{\gamma^\rho (\not{p}' - \not{k}) \gamma^\sigma V_{\rho\sigma}^\mu(q, k - p', p - k)}{k^2 (p' - k)^2 (p - k)^2}, \tag{3.499}$$

where $V_{\rho\sigma}^\mu$ is the three-gauge-boson vertex tensor. The same logic gives the coefficient of the logarithmic divergence. In the ultraviolet part, the three-gauge-boson vertex contributes the part linear in the loop momentum,

$$V_{\rho\sigma}^\mu(q, k - p', p - k) \longrightarrow 2k^\mu g_{\rho\sigma} - k_\rho \delta_\sigma^\mu - k_\sigma \delta_\rho^\mu + \text{terms independent of } k. \tag{3.500}$$

Keeping only the even powers of k after multiplication by $\gamma^\rho (\not{p}' - \not{k}) \gamma^\sigma$, we get

$$\begin{aligned}
\gamma^\rho (\not{p}' - \not{k}) \gamma^\sigma V_{\rho\sigma}^\mu &\longrightarrow -\gamma^\rho \not{k} \gamma^\sigma (2k^\mu g_{\rho\sigma} - k_\rho \delta_\sigma^\mu - k_\sigma \delta_\rho^\mu) \\
&= -2k^\mu k_\alpha \gamma^\rho \gamma^\alpha \gamma_\rho + k_\rho k_\alpha \gamma^\rho \gamma^\alpha \gamma^\mu + k_\sigma k_\alpha \gamma^\mu \gamma^\alpha \gamma^\sigma.
\end{aligned} \tag{3.501}$$

Now use symmetric integration:

$$k^\mu k_\alpha \rightarrow \frac{1}{d} \delta_\alpha^\mu k^2, \quad k_\rho k_\alpha \rightarrow \frac{1}{d} g_{\rho\alpha} k^2, \quad k_\sigma k_\alpha \rightarrow \frac{1}{d} g_{\sigma\alpha} k^2. \tag{3.502}$$

Then

$$\begin{aligned}
\gamma^\rho(\not{p}' - \not{k})\gamma^\sigma V^\mu_{\rho\sigma} &\longrightarrow \frac{k^2}{d} [-2\gamma^\rho\gamma^\mu\gamma_\rho + \gamma^\rho\gamma_\rho\gamma^\mu + \gamma^\mu\gamma^\sigma\gamma_\sigma] \\
&= \frac{k^2}{d} [-2(2-d) + d + d] \gamma^\mu \\
&= \frac{4(d-1)}{d} k^2 \gamma^\mu = 3k^2 \gamma^\mu + \mathcal{O}(\epsilon).
\end{aligned} \tag{3.503}$$

The same logarithmic integral then appears:

$$\int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2)^2} \longrightarrow \frac{i}{(4\pi)^2} \Gamma(\epsilon) (-q^2)^{-\epsilon}. \tag{3.504}$$

Together with the color factor $\frac{1}{2}C_2(G)t^a$ obtained above, this gives

$$\Lambda_{\text{NA,div}}^\mu = ig\gamma^\mu(t^a)_{ij} \frac{g^2}{(4\pi)^2} \frac{3}{2} C_2(G) \Gamma(\epsilon) (-q^2)^{-\epsilon}. \tag{3.505}$$

Counterterm and subtraction

The vertex counterterm comes from $g\delta_1 A_\mu^a \bar{\psi}\gamma^\mu t^a \psi$, hence

$$\Lambda_{\text{ct}}^\mu = ig\gamma^\mu(t^a)_{ij}\delta_1. \tag{3.506}$$

Adding the three pieces, and notice that only the coefficient of $\gamma^\mu(t^a)_{ij}$ is needed for Z_1 .

$$\begin{aligned}
\Lambda_{\text{div}}^\mu &= ig\gamma^\mu(t^a)_{ij} \left[\frac{g^2}{(4\pi)^2} \left(C_2(r) - \frac{1}{2}C_2(G) + \frac{3}{2}C_2(G) \right) \Gamma(\epsilon) (-q^2)^{-\epsilon} + \delta_1 \right] \\
&= ig\gamma^\mu(t^a)_{ij} \left[\frac{g^2}{(4\pi)^2} (C_2(r) + C_2(G)) \Gamma(\epsilon) (-q^2)^{-\epsilon} + \delta_1 \right].
\end{aligned} \tag{3.507}$$

At the subtraction point $-q^2 = \mu^2$, the divergent part is cancelled by choosing

$$0 = \frac{g^2}{(4\pi)^2} (C_2(r) + C_2(G)) \Gamma(\epsilon) (\mu^2)^{-\epsilon} + \delta_1. \tag{3.508}$$

Therefore

$$\boxed{\delta_1 = -\frac{g^2}{(4\pi)^2} (C_2(r) + C_2(G)) \Gamma(\epsilon) (\mu^2)^{-\epsilon}}. \tag{3.509}$$

4.4 Callan–Symanzik equation in non-Abelian gauge theory

Now we can try to write down the Callan–Symanzik equation and analyse the running coupling. For the fermion–gauge-boson three-point Green’s function define

$$G_{A\bar{\psi}\psi} = \langle A_\mu^a(x) \psi_i(y) \bar{\psi}_j(z) \rangle. \tag{3.510}$$

The bare and renormalized fields are related by

$$A_{0,\mu}^a = Z_3^{1/2} A_\mu^a, \quad \psi_0 = Z_2^{1/2} \psi, \quad \bar{\psi}_0 = Z_2^{1/2} \bar{\psi}. \tag{3.511}$$

Therefore the bare three-point Green's function is

$$\begin{aligned} G_{0,A\bar{\psi}\psi} &= \langle A_{0,\mu}^a \psi_{0,i} \bar{\psi}_{0,j} \rangle \\ &= Z_3^{1/2} Z_2 G_{A\bar{\psi}\psi}. \end{aligned} \quad (3.512)$$

Here the factor Z_2 appears because there are two fermion fields, and the factor $Z_3^{1/2}$ appears because there is one gauge field.

Now act with $\mu d/d\mu$ while keeping all bare quantities fixed, and μ is the arbitrary subtraction scale. Since $G_{0,A\bar{\psi}\psi}$ is a bare quantity,

$$\mu \frac{d}{d\mu} G_{0,A\bar{\psi}\psi} = 0. \quad (3.513)$$

Using $G_{0,A\bar{\psi}\psi} = Z_2 Z_3^{1/2} G_{A\bar{\psi}\psi}$, this gives

$$\begin{aligned} 0 &= \mu \frac{d}{d\mu} \left(Z_2 Z_3^{1/2} G_{A\bar{\psi}\psi} \right) \\ &= Z_2 Z_3^{1/2} \left[\mu \frac{d}{d\mu} + \mu \frac{d \ln Z_2}{d\mu} + \frac{1}{2} \mu \frac{d \ln Z_3}{d\mu} \right] G_{A\bar{\psi}\psi}. \end{aligned} \quad (3.514)$$

The total μ derivative acting on the renormalized Green's function contains the explicit scale dependence and the implicit dependence through the running coupling:

$$\mu \frac{d}{d\mu} = \mu \frac{\partial}{\partial \mu} + \mu \frac{dg}{d\mu} \frac{\partial}{\partial g} = \mu \frac{\partial}{\partial \mu} + \beta(g) \frac{\partial}{\partial g}. \quad (3.515)$$

We define the field anomalous dimensions by

$$\gamma_2 = \frac{1}{2} \mu \frac{d \ln Z_2}{d\mu}, \quad \gamma_3 = \frac{1}{2} \mu \frac{d \ln Z_3}{d\mu}. \quad (3.516)$$

Then

$$\mu \frac{d \ln Z_2}{d\mu} = 2\gamma_2, \quad \frac{1}{2} \mu \frac{d \ln Z_3}{d\mu} = \gamma_3. \quad (3.517)$$

Thus the Callan–Symanzik equation for this three-point function is

$$\left(\mu \frac{\partial}{\partial \mu} + \beta(g) \frac{\partial}{\partial g} + 2\gamma_2 + \gamma_3 \right) G_{A\bar{\psi}\psi} = 0. \quad (3.518)$$

In a massive theory one should also add the mass-running term $-\gamma_m m \partial/\partial m$. Here we focus on the massless or high-energy part relevant for the gauge coupling running.

Anomalous dimensions.

At one loop,

$$Z_i = 1 + \delta_i, \quad \ln Z_i = \delta_i + \mathcal{O}(g^4). \quad (3.519)$$

Therefore

$$\gamma_i = \frac{1}{2} \mu \frac{d\delta_i}{d\mu} + \mathcal{O}(g^4). \quad (3.520)$$

The divergent factors found above all have the form

$$F_\epsilon(\mu) = \Gamma(\epsilon)(\mu^2)^{-\epsilon}. \quad (3.521)$$

Since

$$\begin{aligned}\mu \frac{d}{d\mu} F_\epsilon(\mu) &= \Gamma(\epsilon) \mu \frac{d}{d\mu} (\mu^2)^{-\epsilon} \\ &= -2\epsilon \Gamma(\epsilon) (\mu^2)^{-\epsilon} \\ &= -2 + \mathcal{O}(\epsilon),\end{aligned}\tag{3.522}$$

we will use

$$\frac{1}{2} \mu \frac{d}{d\mu} F_\epsilon(\mu) = -1 + \mathcal{O}(\epsilon).\tag{3.523}$$

From the one-loop self-energy and vacuum-polarization counterterms,

$$\begin{aligned}\delta_2 &= -\frac{g^2}{(4\pi)^2} C_2(r) F_\epsilon(\mu), \\ \delta_3 &= \frac{g^2}{(4\pi)^2} \left(\frac{5}{3} C_2(G) - \frac{4}{3} n_f C(r) \right) F_\epsilon(\mu).\end{aligned}\tag{3.524}$$

Thus

$$\begin{aligned}\gamma_2 &= \frac{1}{2} \mu \frac{d\delta_2}{d\mu} = -\frac{g^2}{(4\pi)^2} C_2(r) \left[\frac{1}{2} \mu \frac{dF_\epsilon}{d\mu} \right] \\ &= \frac{g^2}{(4\pi)^2} C_2(r),\end{aligned}\tag{3.525}$$

$$\gamma_3 = \frac{1}{2} \mu \frac{d\delta_3}{d\mu} = -\frac{g^2}{(4\pi)^2} \left(\frac{5}{3} C_2(G) - \frac{4}{3} n_f C(r) \right).\tag{3.526}$$

The sign of γ_3 here follows from the convention $F_\epsilon(\mu) = \Gamma(\epsilon)(\mu^2)^{-\epsilon}$. Some books absorb the opposite sign into the definition of the subtraction factor; then the displayed intermediate sign changes, while the beta function below is unchanged.

Gauge coupling running.

The gauge coupling as we computed in (3.396) at 1-loop level is

$$\delta_g = \delta_1 - \delta_2 - \frac{1}{2} \delta_3,\tag{3.527}$$

where $g_0 = Z_2^{-1} Z_3^{-1/2} Z_1 g$. Because g_0 is bare, it is independent of μ :

$$\begin{aligned}0 &= \mu \frac{dg_0}{d\mu} = \mu \frac{d}{d\mu} (Z_g g) \\ &= Z_g \beta(g) + g \mu \frac{dZ_g}{d\mu}.\end{aligned}\tag{3.528}$$

To one-loop order $Z_g = 1 + \delta_g + \mathcal{O}(g^2)$, so

$$\beta(g) = -g \mu \frac{d\delta_g}{d\mu} + \mathcal{O}(g^5).\tag{3.529}$$

Now insert the one-loop counterterms

$$\begin{aligned}\delta_1 &= -\frac{g^2}{(4\pi)^2} (C_2(r) + C_2(G)) F_\epsilon(\mu), \\ \delta_2 &= -\frac{g^2}{(4\pi)^2} C_2(r) F_\epsilon(\mu), \\ \delta_3 &= \frac{g^2}{(4\pi)^2} \left(\frac{5}{3} C_2(G) - \frac{4}{3} n_f C(r) \right) F_\epsilon(\mu).\end{aligned}\tag{3.530}$$

Then

$$\begin{aligned}\delta_g &= -\frac{g^2}{(4\pi)^2} \left[(C_2(r) + C_2(G)) - C_2(r) + \frac{1}{2} \left(\frac{5}{3} C_2(G) - \frac{4}{3} n_f C(r) \right) \right] F_\epsilon(\mu) \\ &= -\frac{g^2}{(4\pi)^2} \left(\frac{11}{6} C_2(G) - \frac{2}{3} n_f C(r) \right) F_\epsilon(\mu) \\ &= -\frac{1}{2} \frac{g^2}{(4\pi)^2} \left(\frac{11}{3} C_2(G) - \frac{4}{3} n_f C(r) \right) F_\epsilon(\mu).\end{aligned}\tag{3.531}$$

Define

$$\beta_0 = \frac{11}{3} C_2(G) - \frac{4}{3} n_f C(r).\tag{3.532}$$

Therefore

$$\delta_g = -\frac{1}{2} \frac{g^2}{(4\pi)^2} \beta_0 F_\epsilon(\mu).\tag{3.533}$$

Taking the scale derivative,

$$\begin{aligned}\mu \frac{d\delta_g}{d\mu} &= -\frac{1}{2} \frac{g^2}{(4\pi)^2} \beta_0 \mu \frac{dF_\epsilon}{d\mu} + \mathcal{O}(g^4) \\ &= -\frac{1}{2} \frac{g^2}{(4\pi)^2} \beta_0 (-2 + \mathcal{O}(\epsilon)) \\ &= \frac{g^2}{(4\pi)^2} \beta_0.\end{aligned}\tag{3.534}$$

Finally,

$$\boxed{\beta(g) = -g \frac{g^2}{(4\pi)^2} \left(\frac{11}{3} C_2(G) - \frac{4}{3} n_f C(r) \right) + \mathcal{O}(g^4)}\tag{3.535}$$

or, equivalently,

$$\beta(g) = -g \left[\frac{g^2}{(4\pi)^2} \beta_0 + \frac{g^4}{(4\pi)^4} \beta_1 + \dots \right].\tag{3.536}$$

The positive value of β_0 for sufficiently small n_f means

$$\mu \frac{dg}{d\mu} < 0,\tag{3.537}$$

so the gauge coupling becomes smaller at higher energy. This is the one-loop statement of asymptotic freedom in non-Abelian gauge theory.

5. Quantum Chromodynamics: QCD

This section organizes the basic physics of quantum chromodynamics (QCD): why the strong interaction needs color, why the coupling becomes weak at high energies, why perturbation theory breaks down in the infrared, and why apparently infrared-divergent partonic cross sections can still lead to finite physical predictions.

5.1 What QCD is

QCD is the non-Abelian gauge theory of the strong interaction. Its gauge group is

$$SU(3)_{\text{color}}. \quad (3.538)$$

The word “color” does not refer to visible color. It is an internal quantum number carried by quarks. A quark field has a color index

$$q_i(x), \quad i = 1, 2, 3, \quad (3.539)$$

and the three basis states are often called red, green, and blue. The gluon field is written as

$$A_\mu^a(x), \quad a = 1, \dots, 8, \quad (3.540)$$

because gluons transform in the adjoint representation of $SU(3)$, whose dimension is $3^2 - 1 = 8$.

The classical QCD Lagrangian is

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \sum_f \bar{q}_f (i\gamma^\mu D_\mu - m_f) q_f, \quad (3.541)$$

where

$$D_\mu = \partial_\mu + ig_s A_\mu^a T^a, \quad (3.542)$$

$$G_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c. \quad (3.543)$$

The matrices T^a are the generators of $SU(3)$ in the fundamental representation. We use the normalization

$$\text{tr}(T^a T^b) = T_F \delta^{ab}, \quad T_F = \frac{1}{2}. \quad (3.544)$$

For $SU(N_c)$, the standard group factors are

$$C_2(G) = N_c, \quad C_2(r) = \frac{N_c^2 - 1}{2N_c}, \quad T_F = \frac{1}{2}. \quad (3.545)$$

For real QCD, $N_c = 3$, so $C_2(G) = 3$ and $C_2(r) = 4/3$.

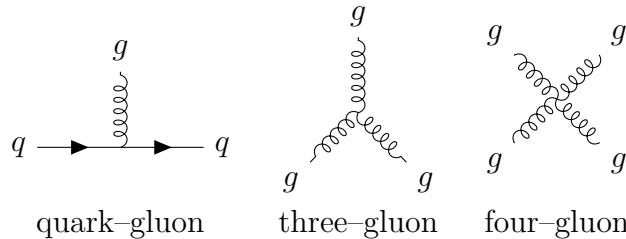


Figure 3.1: The elementary QCD interactions drawn from the QCD Lagrangian. The last two vertices are absent in QED and are responsible for much of the non-Abelian physics of QCD.

5.2 Why color is needed

Color is not an artificial decoration. It is required by both theory and experiment. Three classic pieces of evidence come from baryon spectroscopy, $\pi^0 \rightarrow \gamma\gamma$, and e^+e^- annihilation.

1. The Δ^{++} baryon and Fermi statistics

The quark content of the Δ^{++} baryon is

$$\Delta^{++} = uuu. \quad (3.546)$$

Its spin is $3/2$. In the state with maximal spin projection, the three u quarks have parallel spins:

$$|s_z = 3/2\rangle = |u \uparrow u \uparrow u \uparrow\rangle. \quad (3.547)$$

Thus the spin part of the wave function is completely symmetric. Since all three quarks are u quarks, the flavour part is also symmetric. The ground-state baryon has orbital angular momentum $L = 0$, so the spatial wave function is symmetric as well.

Quarks are fermions. The total wave function must be antisymmetric under the exchange of any two quarks:

$$P_{ij}\Psi_{\text{total}} = -\Psi_{\text{total}}. \quad (3.548)$$

Writing

$$\Psi_{\text{total}} = \Psi_{\text{space}} \Psi_{\text{spin}} \Psi_{\text{flavour}} \Psi_{\text{color}}, \quad (3.549)$$

the first three factors are symmetric for the ground-state Δ^{++} . Therefore the missing antisymmetry must come from the color wave function. The color-singlet baryon wave function is

$$\Psi_{\text{color}} = \frac{1}{\sqrt{6}} \epsilon_{ijk} |i j k\rangle, \quad i, j, k \in \{r, g, b\}. \quad (3.550)$$

For example, exchanging the first two color labels gives

$$\epsilon_{jik} = -\epsilon_{ijk}, \quad (3.551)$$

so the color wave function changes sign. With color included, the full Δ^{++} wave function obeys the Pauli principle. Without color, the observed Δ^{++} state would be inconsistent with Fermi statistics.

2. The decay $\pi^0 \rightarrow \gamma\gamma$

The flavour wave function of the neutral pion is

$$|\pi^0\rangle = \frac{1}{\sqrt{2}} (|u\bar{u}\rangle - |d\bar{d}\rangle). \quad (3.552)$$

The two photons couple to the quarks through a triangle diagram. Since photons couple to

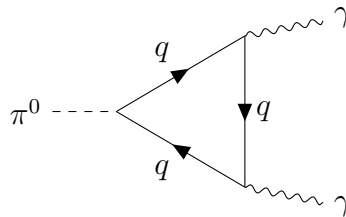


Figure 3.2: Triangle topology for $\pi^0 \rightarrow \gamma\gamma$. The loop is summed over quark flavours and colors.

electric charge, each color gives the same contribution, and summing over color gives a factor of N_c . Up to a common kinematic factor, the charge part of the amplitude is

$$\begin{aligned}\mathcal{A}(\pi^0 \rightarrow \gamma\gamma) &\propto \frac{N_c}{\sqrt{2}} (Q_u^2 - Q_d^2) \\ &= \frac{N_c}{\sqrt{2}} \left[\left(\frac{2}{3}\right)^2 - \left(-\frac{1}{3}\right)^2 \right] = \frac{N_c}{3\sqrt{2}}.\end{aligned}\quad (3.553)$$

More completely, the chiral anomaly gives

$$\Gamma(\pi^0 \rightarrow \gamma\gamma) = \frac{m_\pi^3 \alpha^2}{64\pi^3 f_\pi^2} [N_c (Q_u^2 - Q_d^2)]^2. \quad (3.554)$$

Since $Q_u = 2/3$ and $Q_d = -1/3$, for $N_c = 3$ the factor in brackets is

$$3 \left(\frac{4}{9} - \frac{1}{9} \right) = 1. \quad (3.555)$$

If there were no color, namely $N_c = 1$, the amplitude would be smaller by a factor of 3, and the decay width would be smaller by a factor of 9. The observed size of $\pi^0 \rightarrow \gamma\gamma$ supports $N_c = 3$.

3. The ratio $R(s)$ in $e^+e^- \rightarrow$ hadrons

Define

$$R(s) = \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)}. \quad (3.556)$$

At leading order, and neglecting the lepton mass,

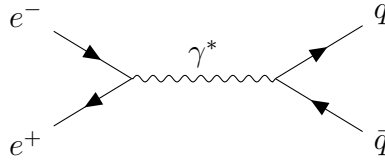


Figure 3.3: Leading-order partonic process behind $e^+e^- \rightarrow$ hadrons. At this order the hadronic final state is approximated by $q\bar{q}$, followed by nonperturbative hadronization.

$$\sigma(e^+e^- \rightarrow \mu^+\mu^-) = \frac{4\pi\alpha^2}{3s}. \quad (3.557)$$

For a final-state quark pair $q\bar{q}$, the photon–quark vertex contains the electric charge Q_q , so the squared amplitude receives a factor Q_q^2 . The final-state quark can also have any of N_c colors, so

$$\sigma(e^+e^- \rightarrow q\bar{q}) = N_c Q_q^2 \sigma(e^+e^- \rightarrow \mu^+\mu^-) \left[1 + \mathcal{O}\left(\frac{m_q^2}{s}\right) \right]. \quad (3.558)$$

Therefore, when the energy is high enough to produce a set of light quark flavours,

$$R(s) = N_c \sum_{q \text{ open}} Q_q^2 + \mathcal{O}(\alpha_s). \quad (3.559)$$

In the energy region where only u, d, s are open,

$$\sum_{q=u,d,s} Q_q^2 = \left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 = \frac{2}{3}. \quad (3.560)$$

Experimentally the corresponding plateau in $R(s)$ is approximately 2, hence

$$2 \simeq N_c \frac{2}{3} \quad \Rightarrow \quad N_c = 3. \quad (3.561)$$

When $\sqrt{s}$ crosses the charm and bottom thresholds, $R(s)$ increases. Near narrow resonances

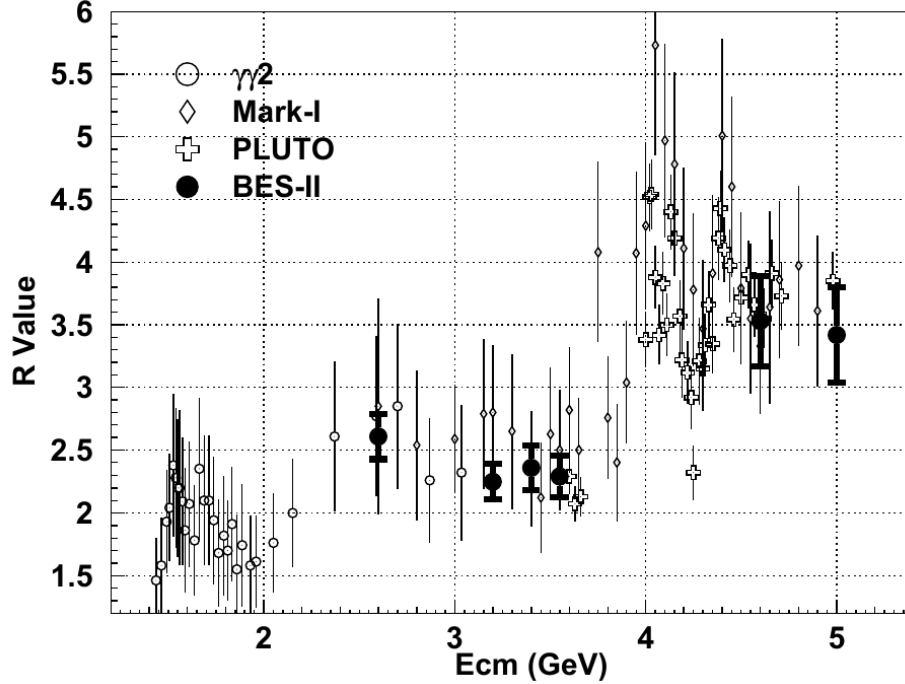


Figure 3.4: Measured values of R in the few-GeV region. The visible step-like structure is the experimental counterpart of changing open quark flavours. Image cropped from BES Collaboration, *Measurements of the cross-section for $e^+e^- \rightarrow \text{hadrons}$ at center-of-mass energies from 2 to 5 GeV*, arXiv:hep-ex/9908046, Fig. 2.

such as J/ψ and Υ , sharp peaks appear.

5.3 No free quarks and confinement

Isolated quarks have not been observed. Quarks and gluons appear only inside color-singlet hadrons. This is called confinement. From the viewpoint of perturbative QCD, confinement cannot be proven by summing a finite number of Feynman diagrams. However, the running of the strong coupling gives an important hint: the coupling is small at short distances and large at long distances, so perturbation theory eventually breaks down in the infrared.

A useful qualitative picture is the static quark potential

$$V(r) \simeq -C_2(r) \frac{\alpha_s(r)}{r} + \sigma r. \quad (3.562)$$

The first term is Coulomb-like at short distances. The linear term at large distances means that separating color charges costs an energy that grows with distance.

5.4 Renormalization constants and the running coupling

The strong coupling is defined by

$$\alpha_s = \frac{g_s^2}{4\pi}. \quad (3.563)$$

In dimensional regularization,

$$d = 4 - 2\epsilon. \quad (3.564)$$

Because g_s has a mass dimension away from four dimensions, the bare coupling is written as

$$g_{s,0} = \mu^\epsilon Z_g g_s(\mu). \quad (3.565)$$

Equivalently, if

$$a_s \equiv \frac{\alpha_s}{4\pi} = \frac{g_s^2}{(4\pi)^2}, \quad (3.566)$$

then

$$a_{s,0} = \mu^{2\epsilon} Z_{a_s} a_s(\mu), \quad Z_{a_s} = Z_g^2. \quad (3.567)$$

At one loop,

$$Z_g = 1 - \frac{1}{2} \frac{g_s^2}{(4\pi)^2} \frac{\beta_0}{\epsilon} + \mathcal{O}(g_s^4), \quad (3.568)$$

where

$$\beta_0 = \frac{11}{3}C_2(G) - \frac{4}{3}T_F n_f = \frac{11}{3}C_2(G) - \frac{2}{3}n_f. \quad (3.569)$$

Here n_f is the number of light quark flavours participating in the running. The first term comes from gluon and ghost effects, especially the non-Abelian self-interaction of gluons. The second term comes from quark loops. Physically, quark loops behave like the screening effect in QED, while gluon self-interactions produce antiscreening. Asymptotic freedom occurs because the gluon contribution dominates.

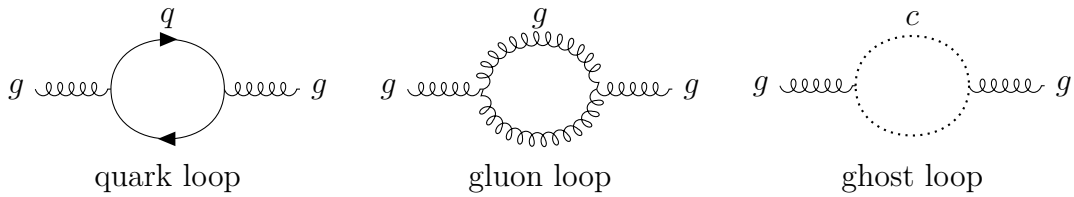


Figure 3.5: One-loop vacuum-polarization topologies contributing to the QCD beta function. The quark loop gives the $-\frac{2}{3}n_f$ part of β_0 for $SU(3)$; the gluon and ghost sectors together give the positive 11 term.

In the four-dimensional limit, the beta function is defined by

$$\frac{da_s}{d \ln \mu^2} = \beta(a_s). \quad (3.570)$$

Its perturbative expansion is

$$\beta(a_s) = -\beta_0 a_s^2 - \beta_1 a_s^3 + \mathcal{O}(a_s^4), \quad (3.571)$$

with

$$\beta_1 = \frac{34}{3}C_2(G)^2 - \frac{20}{3}C_2(G)T_F n_f - 4C_2(r)T_F n_f. \quad (3.572)$$

For $SU(3)$,

$$\beta_0 = 11 - \frac{2}{3}n_f, \quad \beta_1 = 102 - \frac{38}{3}n_f. \quad (3.573)$$

If one uses α_s rather than $a_s = \alpha_s/(4\pi)$, then

$$\frac{d\alpha_s}{d \ln \mu^2} = -\frac{\beta_0}{4\pi}\alpha_s^2 - \frac{\beta_1}{(4\pi)^2}\alpha_s^3 + \dots. \quad (3.574)$$

5.5 Solving the one-loop running explicitly

Keeping only the one-loop term gives

$$\frac{da_s}{d \ln Q^2} = -\beta_0 a_s^2. \quad (3.575)$$

Separate variables:

$$\frac{da_s}{a_s^2} = -\beta_0 d \ln Q^2. \quad (3.576)$$

Integrating from a reference scale μ to a scale Q gives

$$\int_{a_s(\mu^2)}^{a_s(Q^2)} \frac{da}{a^2} = -\beta_0 \int_{\ln \mu^2}^{\ln Q^2} dL, \quad (3.577)$$

$$-\frac{1}{a_s(Q^2)} + \frac{1}{a_s(\mu^2)} = -\beta_0 \ln \frac{Q^2}{\mu^2}. \quad (3.578)$$

Therefore

$$\frac{1}{a_s(Q^2)} = \frac{1}{a_s(\mu^2)} + \beta_0 \ln \frac{Q^2}{\mu^2}. \quad (3.579)$$

Writing the integration constant as Λ_{QCD} , one obtains

$$a_s(Q^2) = \frac{1}{\beta_0 \ln(Q^2/\Lambda_{\text{QCD}}^2)}. \quad (3.580)$$

Thus

$$\boxed{\alpha_s(Q^2) = \frac{4\pi}{\beta_0 \ln(Q^2/\Lambda_{\text{QCD}}^2)}} \quad \text{one-loop.} \quad (3.581)$$

If an experimental input $\alpha_s(\mu^2)$ is given, then

$$\Lambda_{\text{QCD}} = \mu \exp \left[-\frac{1}{2\beta_0 a_s(\mu^2)} \right] = \mu \exp \left[-\frac{2\pi}{\beta_0 \alpha_s(\mu^2)} \right]. \quad (3.582)$$

In precision applications, one uses higher-loop running and threshold matching. A common reference input is

$$\alpha_s(M_Z) \simeq 0.1180. \quad (3.583)$$

For large Q ,

$$\ln \frac{Q^2}{\Lambda_{\text{QCD}}^2} \rightarrow \infty, \quad \alpha_s(Q^2) \rightarrow 0. \quad (3.584)$$

This is asymptotic freedom: high-energy, short-distance processes are weakly coupled and can be treated perturbatively. When Q approaches Λ_{QCD} , the one-loop expression has a Landau-pole-like singularity. This does not mean that a physical observable literally diverges; it means

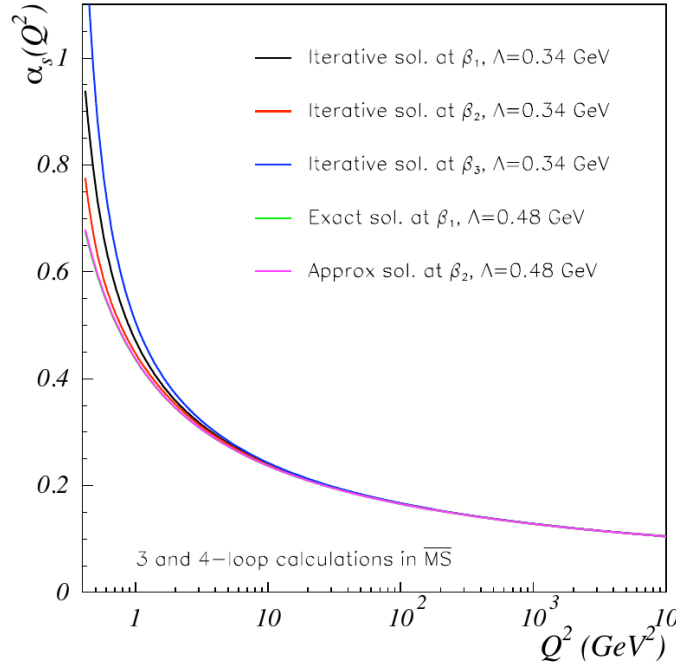


Figure 3.6: Perturbative running of $\alpha_s(Q^2)$ at different loop orders. The coupling decreases at large momentum transfer. Image cropped from A. Deur, S. J. Brodsky, and G. F. de Téramond, *The QCD Running Coupling*, arXiv:1604.08082, Fig. 3.1.

that perturbation theory is no longer reliable and nonperturbative confinement physics becomes important.

Asymptotic freedom requires

$$\beta_0 > 0. \quad (3.585)$$

For $SU(3)$,

$$11 - \frac{2}{3}n_f > 0 \quad \implies \quad n_f < \frac{33}{2} = 16.5. \quad (3.586)$$

Thus integer flavour numbers obeying

$$n_f \leq 16 \quad (3.587)$$

lead to one-loop asymptotic freedom. The real world has at most six quark flavours, so real QCD is safely in the asymptotically free regime.

5.6 The basic idea of a conformal window

Including the two-loop term,

$$\beta(a_s) = -\beta_0 a_s^2 - \beta_1 a_s^3 + \dots \quad (3.588)$$

A nonzero fixed point satisfies

$$\beta(a_s^*) = 0 \quad \implies \quad a_s^* = -\frac{\beta_0}{\beta_1}. \quad (3.589)$$

For this fixed point to occur at positive coupling, β_0 and β_1 must have opposite signs. For $SU(3)$,

$$\beta_0 > 0 \quad \iff \quad n_f < 16.5, \quad (3.590)$$

while

$$\beta_1 < 0 \iff 102 - \frac{38}{3}n_f < 0 \iff n_f > \frac{306}{38} \simeq 8.05. \quad (3.591)$$

Therefore two-loop perturbation theory suggests that, roughly for $9 \leq n_f \leq 16$, an infrared fixed point may exist. When n_f is close to 16.5, β_0 is small and a_s^* is also small. This is the Banks–Zaks fixed point, which is perturbatively controlled. The exact lower edge of the conformal window is a nonperturbative question and cannot be fixed reliably by the two-loop formula alone.

5.7 Classical running in dimensional regularization

In dimensional regularization, the coupling already has a mass dimension. This can be seen from the interaction term

$$g_s \bar{q} \gamma^\mu A_\mu q. \quad (3.592)$$

The field dimensions in d dimensions are

$$[q] = \frac{d-1}{2}, \quad [A_\mu] = \frac{d-2}{2}. \quad (3.593)$$

Since the Lagrangian density has mass dimension d ,

$$[g_s] + 2\frac{d-1}{2} + \frac{d-2}{2} = d, \quad (3.594)$$

$$[g_s] = \frac{4-d}{2} = \epsilon. \quad (3.595)$$

Therefore a dimensionless renormalized coupling requires the factor μ^ϵ :

$$g_{s,0} = \mu^\epsilon Z_g g_s(\mu). \quad (3.596)$$

This produces a classical contribution to the d -dimensional beta function:

$$\boxed{\frac{da_s}{d \ln \mu^2} = -\epsilon a_s - \beta_0 a_s^2 - \beta_1 a_s^3 + \dots} \quad (3.597)$$

The term $-\epsilon a_s$ is not a quantum loop effect. It comes from the engineering dimension of the coupling. Only after taking $\epsilon \rightarrow 0$ does one recover the usual four-dimensional QCD running.

5.8 Heavy-quark thresholds and matching of the running coupling

The number n_f in the running coupling is the number of quark flavours that are effectively light at the scale Q . For example,

$$Q \lesssim m_c : \quad n_f = 3 \quad (u, d, s), \quad (3.598)$$

while

$$m_c \lesssim Q \lesssim m_b : \quad n_f = 4 \quad (u, d, s, c). \quad (3.599)$$

When the scale crosses a heavy-quark mass m_h , one must match the coupling in the n_f -flavour effective theory to the coupling in the $(n_f + 1)$ -flavour theory.

At one loop, the matching is determined by the heavy-quark loop contribution to the gluon propagator. Gauge invariance forces the gluon vacuum polarization to be transverse:

$$i\Pi_{\mu\nu}^{ab}(q) = i\delta^{ab} (g_{\mu\nu} q^2 - q_\mu q_\nu) \Pi(q^2). \quad (3.600)$$

First consider n_f massless light quarks. Their loop gives

$$i\Pi_{\mu\nu}^{ab}(q) = -(-1)g_s^2 T_F n_f \delta^{ab} \mu^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{\text{tr} [\gamma_\mu \not{k} \gamma_\nu (\not{k} + \not{q})]}{k^2 (k+q)^2}. \quad (3.601)$$

The fermion trace is

$$\text{tr} [\gamma_\mu \not{k} \gamma_\nu (\not{k} + \not{q})] = 4 [k_\mu (k+q)_\nu + k_\nu (k+q)_\mu - g_{\mu\nu} k \cdot (k+q)]. \quad (3.602)$$

Introduce a Feynman parameter:

$$\frac{1}{k^2 (k+q)^2} = \int_0^1 dx \frac{1}{[(k+xq)^2 + x(1-x)(-q^2)]^2}. \quad (3.603)$$

After shifting $\ell = k + xq$, all integrals odd in ℓ vanish. Even tensor integrals reduce according to

$$\int d^d \ell \ell_\mu \ell_\nu F(\ell^2) = \frac{g_{\mu\nu}}{d} \int d^d \ell \ell^2 F(\ell^2). \quad (3.604)$$

Only the transverse structure remains. The divergent and logarithmic part of the scalar polarization is

$$\Pi_{\text{light}}^{(n_f)}(q^2) = -\frac{\alpha_s}{4\pi} \frac{4}{3} T_F n_f \left[\frac{1}{\bar{\epsilon}} + \ln \frac{\mu^2}{-q^2} + \text{finite const.} \right], \quad (3.605)$$

or, using $T_F = 1/2$,

$$\Pi_{\text{light}}^{(n_f)}(q^2) = -\frac{\alpha_s}{4\pi} \frac{2}{3} n_f \left[\frac{1}{\bar{\epsilon}} + \ln \frac{\mu^2}{-q^2} + \text{finite const.} \right]. \quad (3.606)$$

Here

$$\frac{1}{\bar{\epsilon}} = \frac{1}{\epsilon} - \gamma_E + \ln 4\pi. \quad (3.607)$$

In handwritten notes one often suppresses scheme-dependent constants and keeps only the pole and logarithm, since these are the terms that determine running and matching.

Now suppose the quark in the loop has mass m_h , and the external momentum is much smaller than that mass:

$$q^2 \ll m_h^2. \quad (3.608)$$

The same Feynman-parameter calculation gives

$$\Pi_h(q^2) = -\frac{\alpha_s}{4\pi} \frac{2}{3} \left[\frac{1}{\bar{\epsilon}} + \ln \frac{\mu^2}{m_h^2} + \mathcal{O}\left(\frac{q^2}{m_h^2}\right) \right]. \quad (3.609)$$

In the $\overline{\text{MS}}$ scheme, after subtracting $1/\bar{\epsilon}$, the heavy-quark loop leaves only a local low-energy correction. It can be absorbed into the definition of the coupling in the low-energy theory. The one-loop decoupling relation is

$$\boxed{\alpha_s^{(n_f)}(\mu) = \alpha_s^{(n_f+1)}(\mu) \left[1 + \frac{\alpha_s^{(n_f+1)}(\mu)}{6\pi} \ln \frac{\mu^2}{m_h^2} \right] + \mathcal{O}(\alpha_s^3)}. \quad (3.610)$$

If the matching scale is chosen to be $\mu = m_h$, then at one loop

$$\alpha_s^{(n_f)}(m_h) = \alpha_s^{(n_f+1)}(m_h) + \mathcal{O}(\alpha_s^3). \quad (3.611)$$

Thus the running coupling is continuous across a heavy-quark threshold at one-loop order. At higher orders finite matching corrections remain, so precision QCD requires threshold matching at each heavy-quark mass.

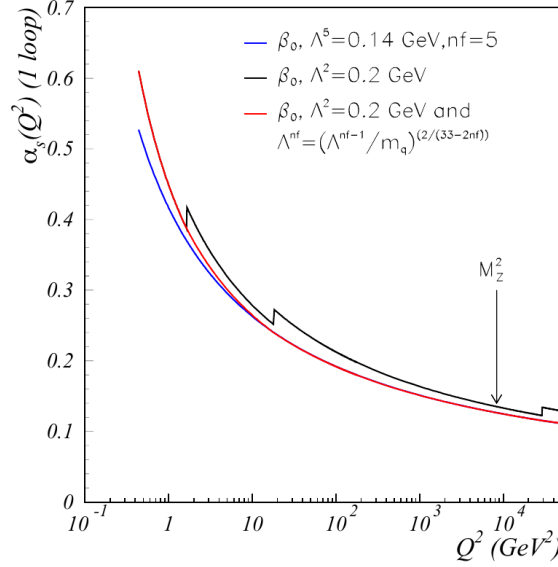


Figure 3.7: Effect of quark thresholds on the running coupling. The red curve illustrates threshold matching, while the black curve shows the discontinuous behaviour that one would obtain from changing the active flavour number without matching. Image cropped from Deur, Brodsky, and de Téramond, arXiv:1604.08082, Fig. 3.2.

5.9 Cross-section predictions and infrared safety

QCD predicts many collider observables, for example

- hadron production in e^+e^- annihilation: LEP, Belle II, BES;
- deep inelastic scattering: HERA, EIC;
- the Drell–Yan process: $pp \rightarrow \gamma^*/Z \rightarrow \ell^+\ell^-$;
- jets, heavy flavour, and vector-boson production at the LHC.

The key requirement is infrared safety: the observable must not change under the emission of a soft gluon or under a collinear splitting.

Consider inclusive $e^+e^- \rightarrow \text{hadrons}$. At leading order,

$$R_0(s) = N_c \sum_q Q_q^2. \quad (3.612)$$

At next-to-leading order there are two classes of corrections:

$$\text{real emission:} \quad e^+e^- \rightarrow q\bar{q}g, \quad (3.613)$$

$$\text{virtual correction:} \quad e^+e^- \rightarrow q\bar{q} \quad \text{with one gluon loop.} \quad (3.614)$$

Each class is infrared divergent by itself. But the inclusive hadronic cross section counts both $q\bar{q}$ and $q\bar{q}g$ final states as hadrons, and the soft and collinear singularities cancel in the sum.

Let a massless quark have momentum

$$p = (E_q, \vec{p}), \quad p^2 = 0, \quad (3.615)$$

and let it radiate a gluon

$$k = (E_g, \vec{k}), \quad k^2 = 0. \quad (3.616)$$

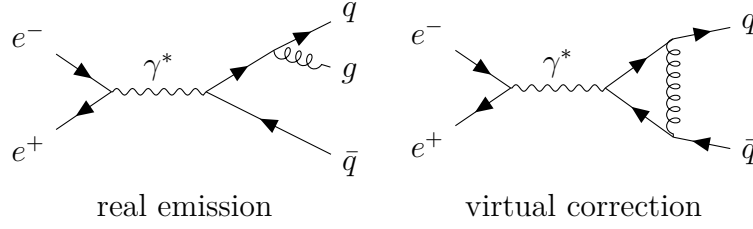


Figure 3.8: The two NLO ingredients in inclusive $e^+e^- \rightarrow \text{hadrons}$. Each piece is infrared divergent separately, but the divergences cancel in the inclusive sum.

If the angle between them is θ_{qg} , then since $|\vec{p}| = E_q$ and $|\vec{k}| = E_g$,

$$2p \cdot k = 2E_q E_g (1 - \cos \theta_{qg}). \quad (3.617)$$

When the gluon is soft,

$$E_g \rightarrow 0, \quad (3.618)$$

this denominator becomes small. When the gluon is collinear to the quark,

$$\theta_{qg} \rightarrow 0, \quad 1 - \cos \theta_{qg} \simeq \frac{\theta_{qg}^2}{2}, \quad (3.619)$$

the denominator also becomes small.

In the soft limit, the squared matrix element factorizes:

$$|\mathcal{M}_{q\bar{q}g}|^2 \simeq |\mathcal{M}_{q\bar{q}}|^2 4\pi\alpha_s C_2(r) \frac{2p_q \cdot p_{\bar{q}}}{(p_q \cdot k)(p_{\bar{q}} \cdot k)}. \quad (3.620)$$

This eikonal factor contains two denominators of the form $p \cdot k$, so it generates soft singularities. In the collinear limit, the propagator

$$(p + k)^2 = p^2 + k^2 + 2p \cdot k = 2p \cdot k \quad (3.621)$$

approaches zero, generating a collinear singularity.

The phase-space measure makes the logarithmic structure explicit. In $d = 4 - 2\epsilon$, a soft gluon has approximately

$$\frac{d^{d-1}k}{2E_g} \sim dE_g E_g^{1-2\epsilon} d\theta \theta^{1-2\epsilon}. \quad (3.622)$$

Multiplying by the soft-collinear eikonal factor

$$\frac{1}{E_g^2 \theta^2}, \quad (3.623)$$

one obtains

$$\int_0^\Lambda dE_g E_g^{-1-2\epsilon} \int_0^\delta d\theta \theta^{-1-2\epsilon}. \quad (3.624)$$

Therefore

$$\int_0^\Lambda dE_g E_g^{-1-2\epsilon} = -\frac{1}{2\epsilon_{\text{IR}}} + \ln \Lambda + \dots, \quad (3.625)$$

$$\int_0^\delta d\theta \theta^{-1-2\epsilon} = -\frac{1}{2\epsilon_{\text{IR}}} + \ln \delta + \dots. \quad (3.626)$$

The energy integral gives a soft pole, and the angular integral gives a collinear pole. Their overlap gives a double pole $1/\epsilon_{\text{IR}}^2$.

The one-loop structure can be summarized as

$$\sigma_{\text{real}} = \sigma_0 \frac{\alpha_s}{4\pi} C_2(r) \left[\frac{2}{\epsilon_{\text{IR}}^2} + \frac{3}{\epsilon_{\text{IR}}} + \text{finite}_{\text{real}} \right], \quad (3.627)$$

$$\sigma_{\text{virtual}} = \sigma_0 \frac{\alpha_s}{4\pi} C_2(r) \left[-\frac{2}{\epsilon_{\text{IR}}^2} - \frac{3}{\epsilon_{\text{IR}}} + \text{finite}_{\text{virtual}} \right]. \quad (3.628)$$

The infrared poles cancel in the sum:

$$\sigma_{\text{NLO}} = \sigma_0 + \sigma_{\text{real}} + \sigma_{\text{virtual}} \quad \text{is finite.} \quad (3.629)$$

This is the KLN theorem at work in the inclusive e^+e^- hadronic cross section.

The finite part gives the famous QCD correction

$$R(s) = N_c \sum_q Q_q^2 \left[1 + \frac{\alpha_s(s)}{4\pi} 3C_2(r) + \mathcal{O}(\alpha_s^2) \right]. \quad (3.630)$$

For QCD, $C_2(r) = 4/3$, so

$$\frac{\alpha_s}{4\pi} 3C_2(r) = \frac{\alpha_s}{4\pi} \cdot 4 = \frac{\alpha_s}{\pi}. \quad (3.631)$$

Thus

$$\boxed{R(s) = N_c \sum_q Q_q^2 \left[1 + \frac{\alpha_s(s)}{\pi} + \mathcal{O}(\alpha_s^2) \right]}. \quad (3.632)$$

This formula ties together the main themes of this section: the number of colors fixes the leading normalization, the coupling $\alpha_s(s)$ runs with the energy scale, and the real and virtual infrared singularities cancel in an inclusive observable.

6. Spontaneous Symmetry Breaking

Spontaneous symmetry breaking is the situation in which the equations and the Lagrangian have a symmetry, but the vacuum chosen by the theory does not. The symmetry is still present in the theory; it is not a mistake in the Lagrangian. The point is that the set of lowest-energy field configurations contains more than one equivalent choice, and choosing one of them hides part of the symmetry.

This section develops the idea in three steps. First we study a real scalar with a discrete symmetry, where there is symmetry breaking but no Goldstone boson. Then we study a complex scalar with a continuous $U(1)$ symmetry, where a massless Goldstone mode appears. Finally we generalize to N complex scalars and prove the Goldstone theorem by counting flat directions of the potential.

6.1 A real scalar with a discrete symmetry

Consider one real scalar field ϕ with Lagrangian

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad (3.633)$$

and potential

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{\lambda}{4}\phi^4, \quad \mu^2 > 0, \quad \lambda > 0. \quad (3.634)$$

This is often called the “wrong-sign mass” potential. Around $\phi = 0$ the quadratic term in the potential has negative curvature, so $\phi = 0$ is not a stable vacuum. This corrects a common misleading phrase: the theory does have a quadratic term; what is wrong is the sign of the quadratic term around the origin.

The potential is invariant under the discrete transformation

$$\phi \longrightarrow -\phi, \quad V(\phi) = V(-\phi). \quad (3.635)$$

This is a $\mathbb{Z}_2$ symmetry.

To find the extrema, compute

$$\frac{dV}{d\phi} = -\mu^2\phi + \lambda\phi^3 = \phi(-\mu^2 + \lambda\phi^2). \quad (3.636)$$

Thus

$$\frac{dV}{d\phi} = 0 \quad \implies \quad \phi = 0 \quad \text{or} \quad \phi^2 = \frac{\mu^2}{\lambda}. \quad (3.637)$$

Define

$$v = \frac{\mu}{\sqrt{\lambda}}. \quad (3.638)$$

The extrema are therefore

$$\phi = 0, \quad \phi = \pm v. \quad (3.639)$$

To decide which are minima, compute the second derivative:

$$\frac{d^2V}{d\phi^2} = -\mu^2 + 3\lambda\phi^2. \quad (3.640)$$

At the origin,

$$\left. \frac{d^2V}{d\phi^2} \right|_{\phi=0} = -\mu^2 < 0, \quad (3.641)$$

so $\phi = 0$ is a local maximum. At $\phi = \pm v$,

$$\left. \frac{d^2V}{d\phi^2} \right|_{\phi=\pm v} = -\mu^2 + 3\lambda\frac{\mu^2}{\lambda} = 2\mu^2 > 0, \quad (3.642)$$

so $\phi = \pm v$ are true minima.

The theory has the symmetry $\phi \rightarrow -\phi$, but the vacuum does not. If the system chooses

$$\langle 0|\phi|0\rangle = +v, \quad (3.643)$$

then the transformed configuration has

$$\langle 0|\phi|0\rangle \longrightarrow -v, \quad (3.644)$$

which is a different vacuum. The two vacua are physically equivalent, but once one is chosen the symmetry is no longer manifest. This is spontaneous breaking of a discrete symmetry.

There is no Goldstone boson in this example. Goldstone bosons arise from broken continuous symmetries, where moving along the vacuum manifold can be done infinitesimally with no energy cost. A discrete symmetry does not provide such a continuous flat direction.

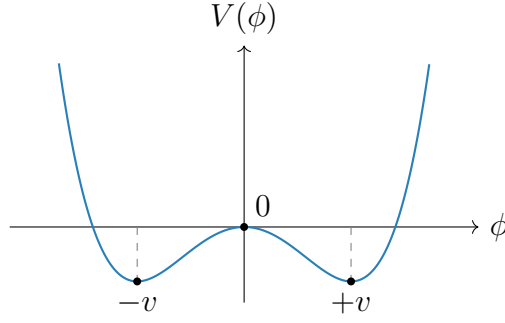


Figure 3.9: The double-well potential. The Lagrangian is symmetric under $\phi \rightarrow -\phi$, but either vacuum $\phi = +v$ or $\phi = -v$ chooses one side.

6.2 A complex scalar and global U(1) breaking

Now consider a complex scalar field

$$\phi = \frac{1}{\sqrt{2}} (\phi_1 + i\phi_2), \quad (3.645)$$

with Lagrangian

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - V(\phi, \phi^*), \quad (3.646)$$

and potential

$$V(\phi, \phi^*) = -\mu^2 \phi^* \phi + \frac{\lambda}{2} (\phi^* \phi)^2, \quad \mu^2 > 0, \quad \lambda > 0. \quad (3.647)$$

The Lagrangian is invariant under the global $U(1)$ transformation

$$\phi(x) \longrightarrow \phi'(x) = e^{i\alpha} \phi(x), \quad \alpha \in \mathbb{R}. \quad (3.648)$$

This is a global symmetry because α is constant in spacetime.

The potential depends only on the radial variable $\phi^* \phi$. Minimizing with respect to ϕ^* gives

$$\frac{\partial V}{\partial \phi^*} = \phi (-\mu^2 + \lambda \phi^* \phi) = 0. \quad (3.649)$$

Therefore

$$\phi = 0 \quad \text{or} \quad \phi^* \phi = \frac{\mu^2}{\lambda}. \quad (3.650)$$

Again $\phi = 0$ is the unstable point. The true vacua satisfy

$$|\phi|^2 = \frac{\mu^2}{\lambda}. \quad (3.651)$$

In the complex plane this is a circle. Since

$$|\phi|^2 = \frac{1}{2} (\phi_1^2 + \phi_2^2), \quad (3.652)$$

the vacuum circle can also be written as

$$\phi_1^2 + \phi_2^2 = \frac{2\mu^2}{\lambda}. \quad (3.653)$$

It is conventional to define

$$v^2 = \frac{2\mu^2}{\lambda}. \quad (3.654)$$

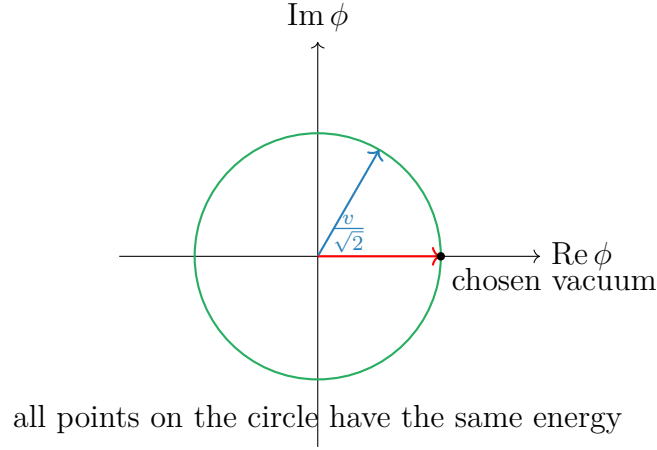


Figure 3.10: The vacuum manifold for the broken global $U(1)$ model. Choosing one point on the circle breaks the manifest $U(1)$ symmetry.

Then the vacuum condition is

$$|\phi|^2 = \frac{v^2}{2}. \quad (3.655)$$

Every point on this circle is a possible ground state. Choosing one point breaks the global $U(1)$ symmetry. We may choose the vacuum to lie along the real direction:

$$\phi_0 = \frac{v}{\sqrt{2}}, \quad \phi_{1,0} = v, \quad \phi_{2,0} = 0. \quad (3.656)$$

This choice is not a loss of generality; any other point on the circle can be rotated into this one by a $U(1)$ transformation.

6.3 Expansion around the chosen vacuum

To find the physical particles, expand the field around the chosen vacuum:

$$\phi(x) = \frac{1}{\sqrt{2}} [v + h(x) + i\sigma(x)]. \quad (3.657)$$

Here $h(x)$ is the radial fluctuation and $\sigma(x)$ is the angular fluctuation. At the vacuum,

$$h = 0, \quad \sigma = 0. \quad (3.658)$$

The kinetic term becomes

$$\begin{aligned} \partial_\mu \phi^* \partial^\mu \phi &= \frac{1}{2} \partial_\mu (v + h - i\sigma) \partial^\mu (v + h + i\sigma) \\ &= \frac{1}{2} \partial_\mu h \partial^\mu h + \frac{1}{2} \partial_\mu \sigma \partial^\mu \sigma. \end{aligned} \quad (3.659)$$

Since v is constant, its derivative vanishes.

Now compute the potential in terms of h and σ . First,

$$\begin{aligned} \phi^* \phi &= \frac{1}{2} [(v + h)^2 + \sigma^2] \\ &= \frac{1}{2} [v^2 + 2vh + h^2 + \sigma^2]. \end{aligned} \quad (3.660)$$

Therefore

$$V(h, \sigma) = -\frac{\mu^2}{2} [(v+h)^2 + \sigma^2] + \frac{\lambda}{8} [(v+h)^2 + \sigma^2]^2. \quad (3.661)$$

Expanding the square,

$$\begin{aligned} [(v+h)^2 + \sigma^2]^2 &= (v^2 + 2vh + h^2 + \sigma^2)^2 \\ &= v^4 + 4v^3h + 6v^2h^2 + 2v^2\sigma^2 + \mathcal{O}(\text{cubic}). \end{aligned} \quad (3.662)$$

Hence the terms up to quadratic order are

$$V(h, \sigma) = -\frac{\mu^2}{2} (v^2 + 2vh + h^2 + \sigma^2) + \frac{\lambda}{8} (v^4 + 4v^3h + 6v^2h^2 + 2v^2\sigma^2) + \dots. \quad (3.663)$$

Separate constant, linear, and quadratic terms:

$$\begin{aligned} V(h, \sigma) &= \left(-\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{8} \right) + \left(-\mu^2 v + \frac{\lambda v^3}{2} \right) h \\ &\quad + \left(-\frac{\mu^2}{2} + \frac{3\lambda v^2}{4} \right) h^2 + \left(-\frac{\mu^2}{2} + \frac{\lambda v^2}{4} \right) \sigma^2 + \dots. \end{aligned} \quad (3.664)$$

Using

$$v^2 = \frac{2\mu^2}{\lambda}, \quad (3.665)$$

the linear term vanishes:

$$-\mu^2 v + \frac{\lambda v^3}{2} = v \left(-\mu^2 + \frac{\lambda v^2}{2} \right) = 0. \quad (3.666)$$

This must happen because we expanded around a minimum. The quadratic terms become

$$-\frac{\mu^2}{2} + \frac{3\lambda v^2}{4} = -\frac{\mu^2}{2} + \frac{3\mu^2}{2} = \mu^2, \quad (3.667)$$

$$-\frac{\mu^2}{2} + \frac{\lambda v^2}{4} = -\frac{\mu^2}{2} + \frac{\mu^2}{2} = 0. \quad (3.668)$$

Thus

$$V(h, \sigma) = V(v) + \mu^2 h^2 + \mathcal{O}(h^3, h^2\sigma, h\sigma^2, \sigma^4). \quad (3.669)$$

The Lagrangian contains $-\mu^2 h^2$. Comparing with the standard mass term

$$-\frac{1}{2} m_h^2 h^2, \quad (3.670)$$

we find

$$m_h^2 = 2\mu^2. \quad (3.671)$$

There is no quadratic term for σ , so

$$m_\sigma^2 = 0. \quad (3.672)$$

The radial field h is massive. The angular field σ is massless and is called the Goldstone boson.

The reason is geometric. The h direction moves away from the vacuum circle, so the potential increases. The σ direction moves along the vacuum circle, where the potential is flat. A flat direction has zero curvature, and zero curvature means zero mass squared.

6.4 N complex scalar fields

Now consider N complex scalar fields ϕ^i , $i = 1, \dots, N$, with

$$\mathcal{L} = \sum_{i=1}^N \partial_\mu \phi^{i*} \partial^\mu \phi^i - V(\phi, \phi^*), \quad (3.673)$$

and

$$V(\phi, \phi^*) = -\mu^2 \sum_{i=1}^N \phi^{i*} \phi^i + \frac{\lambda}{2} \left(\sum_{i=1}^N \phi^{i*} \phi^i \right)^2. \quad (3.674)$$

This potential is invariant under global $U(N)$ transformations:

$$\phi^i \longrightarrow \phi'^i = \sum_{j=1}^N U^{ij} \phi^j, \quad U^\dagger U = \mathbf{1}. \quad (3.675)$$

The quantity

$$\sum_i \phi^{i*} \phi^i \quad (3.676)$$

is just the squared length of the complex vector ϕ , so it is unchanged by unitary rotations.

The minimum satisfies

$$\sum_{i=1}^N \phi^{i*} \phi^i = \frac{\mu^2}{\lambda}. \quad (3.677)$$

This is a sphere in the $2N$ real-dimensional field space. Define again

$$\frac{v^2}{2} = \frac{\mu^2}{\lambda}, \quad v^2 = \frac{2\mu^2}{\lambda}. \quad (3.678)$$

Using the $U(N)$ symmetry, choose the vacuum to point in the first complex direction:

$$\phi_0^1 = \frac{v}{\sqrt{2}}, \quad \phi_0^2 = \dots = \phi_0^N = 0. \quad (3.679)$$

Then write the fields as

$$\phi^1 = \frac{1}{\sqrt{2}} [v + h(x) + i\sigma(x)], \quad (3.680)$$

$$\phi^k = \frac{1}{\sqrt{2}} [\pi_{2k-2}(x) + i\pi_{2k-1}(x)], \quad k = 2, \dots, N. \quad (3.681)$$

There are $2N$ real fields in total:

$$h, \quad \sigma, \quad \pi_2, \dots, \pi_{2N-1}. \quad (3.682)$$

The radial field h changes the length of the vector ϕ and is massive. All other fields change only the direction of ϕ in the vacuum manifold and are massless at tree level.

Equivalently, the chosen vacuum leaves a residual $U(N-1)$ symmetry acting on the fields

$$\phi^2, \dots, \phi^N. \quad (3.683)$$

Thus the symmetry-breaking pattern is

$$U(N) \longrightarrow U(N-1). \quad (3.684)$$

The number of broken generators is

$$\dim U(N) - \dim U(N-1) = N^2 - (N-1)^2 = 2N-1. \quad (3.685)$$

Therefore there are

$$2N-1 \quad (3.686)$$

massless Goldstone bosons. This agrees with the field decomposition above: σ plus the $2N-2$ real components of $\phi^2, \dots, \phi^N$.

6.5 Goldstone theorem

The examples above illustrate the general result.

Goldstone theorem. In a relativistic theory with a continuous global internal symmetry, each spontaneously broken generator gives one massless real scalar excitation, called a Goldstone boson.

There are important assumptions in this statement. The symmetry must be global, not gauged. If the symmetry is gauged, the massless scalar is removed from the physical spectrum by the Higgs mechanism. Also, the symmetry must be exact. If it is explicitly broken, the would-be Goldstone boson generally becomes light but not massless.

6.6 Proof by the mass matrix

Consider N real scalar fields ϕ^i with Lagrangian

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \partial_\mu \phi^i \partial^\mu \phi^i - V(\phi). \quad (3.687)$$

Suppose the potential is invariant under a continuous symmetry group G . For an infinitesimal transformation,

$$\phi^i \longrightarrow \phi^i + \delta_a \phi^i, \quad (3.688)$$

where

$$\delta_a \phi^i = \alpha^a (T_a)^i_j \phi^j. \quad (3.689)$$

Here T_a is a generator of the symmetry and α^a is an infinitesimal real parameter. Since V is invariant,

$$\delta_a V = \sum_i \frac{\partial V}{\partial \phi^i} \delta_a \phi^i = 0. \quad (3.690)$$

Substitute the infinitesimal transformation:

$$\sum_i \frac{\partial V}{\partial \phi^i} (T_a)^i_j \phi^j = 0. \quad (3.691)$$

This identity holds for every field configuration ϕ , not only at the minimum.

Now differentiate this identity with respect to ϕ^k :

$$\sum_i \frac{\partial^2 V}{\partial \phi^k \partial \phi^i} (T_a)^i_j \phi^j + \sum_i \frac{\partial V}{\partial \phi^i} (T_a)^i_k = 0. \quad (3.692)$$

Evaluate at a vacuum ϕ_0 . Since ϕ_0 is a minimum,

$$\left. \frac{\partial V}{\partial \phi^i} \right|_{\phi_0} = 0. \quad (3.693)$$

The second term therefore vanishes. Define the mass matrix

$$(M^2)_{ki} = \left. \frac{\partial^2 V}{\partial \phi^k \partial \phi^i} \right|_{\phi_0}. \quad (3.694)$$

Then

$$\sum_i (M^2)_{ki} (T_a)^i{}_j \phi_0^j = 0. \quad (3.695)$$

In vector notation this is

$$M^2 (T_a \phi_0) = 0. \quad (3.696)$$

Now distinguish two cases.

Unbroken generator.

If T_a is unbroken, it leaves the vacuum invariant:

$$T_a \phi_0 = 0. \quad (3.697)$$

Then the equation above is trivial and gives no new massless field.

Broken generator.

If T_a is broken, then it moves the vacuum:

$$T_a \phi_0 \neq 0. \quad (3.698)$$

But the mass-matrix equation says

$$M^2 (T_a \phi_0) = 0. \quad (3.699)$$

Thus $T_a \phi_0$ is a nonzero eigenvector of the mass matrix with eigenvalue zero. A zero eigenvalue of the scalar mass matrix means a massless scalar mode. Therefore each broken generator gives a massless scalar field.

This is the mathematical form of the geometric picture. Broken generators move the vacuum along the manifold of degenerate minima. Motion along that manifold costs no potential energy, so the corresponding direction has zero curvature. Zero curvature of the potential is precisely zero mass squared.